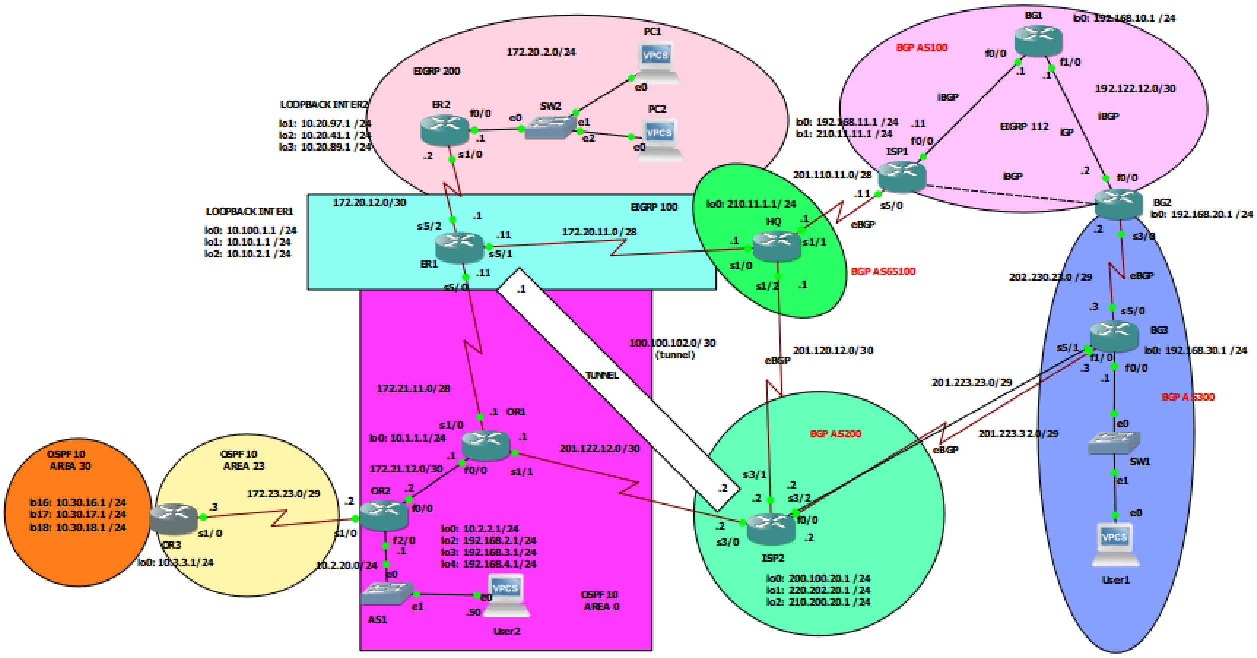
ADVANCED IP ROUTING PROJECT

Write synopsis of this case study and screenshot the topology in simulation

This case study focuses on the implementation and troubleshooting of IP routing in an enterprise network using multiple routing protocols such as EIGRP, OSPF, and BGP. The network is designed to simulate a real-world environment where different autonomous systems must interconnect and share routes efficiently. Key tasks include configuring basic router settings, setting up dynamic routing protocols (EIGRP and OSPF), implementing external BGP (eBGP) and internal BGP (iBGP), route redistribution, and summarization. Additional advanced configurations involve GRE tunneling for alternate routing paths, DHCP services, and Policy-Based Routing (PBR) to manipulate traffic flow. Students are also required to apply route filtering and path manipulation using BGP attributes such as local preference. The main objective is to ensure seamless end-to-end connectivity between users across domains, optimize routing paths, and provide network resilience and redundancy. The case study is carried out using GNS3 simulation software and emphasizes leadership, autonomy, and technical troubleshooting skills in a multi-protocol routing environment.



CONFIGURATION COMMAND

#1: Basic Configuration - `#show ip interface brief`

Output/Result: (You may snapshot from the simulation-GNS3 or PT).) Use the verification command.

HQ

```
HQ#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES NVRAM  administratively down down
Serial1/0              172.20.11.1     YES NVRAM  up                    up
Serial1/1              201.110.11.1    YES NVRAM  up                    up
Serial1/2              201.120.12.1    YES NVRAM  up                    up
Serial1/3              unassigned      YES NVRAM  administratively down down
FastEthernet2/0        unassigned      YES NVRAM  administratively down down
FastEthernet2/1        unassigned      YES NVRAM  administratively down down
Loopback0              210.11.1.1      YES NVRAM  up                    up
```

ER2 (

```
ER2#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        172.20.2.1      YES NVRAM  up                    up
Serial1/0              172.20.12.2     YES NVRAM  up                    up
Serial1/1              unassigned      YES NVRAM  administratively down down
Serial1/2              unassigned      YES NVRAM  administratively down down
Serial1/3              unassigned      YES NVRAM  administratively down down
Loopback1              10.20.97.1      YES NVRAM  up                    up
Loopback2              10.20.41.1      YES NVRAM  up                    up
Loopback3              10.20.89.1      YES NVRAM  up                    up
```

OR1

```
OR1#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        172.21.12.1     YES NVRAM  up                    up
Serial1/0              172.21.11.1     YES NVRAM  up                    up
Serial1/1              201.122.12.1    YES NVRAM  up                    up
Serial1/2              unassigned      YES NVRAM  administratively down down
Serial1/3              unassigned      YES NVRAM  administratively down down
GigabitEthernet3/0     unassigned      YES NVRAM  administratively down down
FastEthernet4/0        unassigned      YES NVRAM  administratively down down
FastEthernet4/1        unassigned      YES NVRAM  administratively down down
Loopback0              10.1.1.1        YES NVRAM  up                    up
```

ISP1

```
ISP1#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.121.11.11   YES NVRAM  up                    up
FastEthernet1/0        unassigned      YES NVRAM  administratively down down
FastEthernet1/1        unassigned      YES NVRAM  administratively down down
FastEthernet2/0        unassigned      YES NVRAM  administratively down down
FastEthernet2/1        unassigned      YES NVRAM  administratively down down
GigabitEthernet4/0     unassigned      YES NVRAM  administratively down down
Serial5/0              201.110.11.11   YES NVRAM  up                    up
Serial5/1              unassigned      YES NVRAM  administratively down down
Serial5/2              unassigned      YES NVRAM  administratively down down
Serial5/3              unassigned      YES NVRAM  administratively down down
Serial6/0              unassigned      YES NVRAM  administratively down down
Serial6/1              unassigned      YES NVRAM  administratively down down
Serial6/2              unassigned      YES NVRAM  administratively down down
Serial6/3              unassigned      YES NVRAM  administratively down down
Loopback0              192.168.11.1    YES NVRAM  up                    up
Loopback1              210.11.11.1     YES NVRAM  up                    up
```

BG1

```
BG1#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.121.11.1    YES NVRAM  up                    up
FastEthernet1/0        192.122.12.1    YES NVRAM  up                    up
FastEthernet1/1        unassigned      YES NVRAM  administratively down down
Loopback0              192.168.10.1    YES NVRAM  up                    up
```

BG3

```
BG3#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        203.23.30.1     YES NVRAM  up                    up
FastEthernet1/0        201.223.23.3    YES NVRAM  up                    up
FastEthernet1/1        unassigned      YES NVRAM  administratively down down
FastEthernet2/0        unassigned      YES NVRAM  administratively down down
FastEthernet2/1        unassigned      YES NVRAM  administratively down down
GigabitEthernet3/0     unassigned      YES NVRAM  administratively down down
GigabitEthernet4/0     unassigned      YES NVRAM  administratively down down
Serial5/0              202.230.23.3    YES NVRAM  up                    up
Serial5/1              201.223.32.3    YES NVRAM  up                    up
Serial5/2              unassigned      YES NVRAM  administratively down down
Serial5/3              unassigned      YES NVRAM  administratively down down
Serial6/0              unassigned      YES NVRAM  administratively down down
Serial6/1              unassigned      YES NVRAM  administratively down down
Serial6/2              unassigned      YES NVRAM  administratively down down
Serial6/3              unassigned      YES NVRAM  administratively down down
Loopback0              192.168.30.1    YES NVRAM  up                    up
```

#2: EIGRP Configuration – `#show ip eigrp neighbour`

Output/Result: (You may snapshot from the simulation-GNS3 or PT).)

HQ `#show ip eigrp neighbour`

```
HQ#sh ip eigrp neigh
EIGRP-IPv4 Neighbors for AS(100)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
0   172.20.11.11            Se1/0                    14 00:00:12  103   618  0  19
```

ER1 `#show ip route`

```
ER1#sh ip route eigrp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 172.21.11.1 to network 0.0.0.0

      10.0.0.0/8 is variably subnetted, 15 subnets, 4 masks
D        10.10.0.0/16 is a summary, 00:25:09, Null0
D        10.20.41.0/24 [90/2297856] via 172.20.12.2, 00:25:08, Serial5/2
D        10.20.89.0/24 [90/2297856] via 172.20.12.2, 00:25:08, Serial5/2
D        10.20.97.0/24 [90/2297856] via 172.20.12.2, 00:25:08, Serial5/2
      172.20.0.0/16 is variably subnetted, 5 subnets, 4 masks
D        172.20.2.0/24 [90/2172416] via 172.20.12.2, 00:25:08, Serial5/2
```

ER2 `#show ip eigrp neighbour & Show ip route`

```
ER2#show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(200)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
0   172.20.12.1             Se1/0                    12 00:05:27   40   240  0  3
EIGRP-IPv4 Neighbors for AS(100)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
0   172.20.12.1             Se1/0                    10 00:01:18   34   204  0  2
```

Explanation

The `show ip eigrp neighbors` command confirms successful EIGRP adjacency between routers by displaying neighbor IDs, interfaces, and uptime. This ensures routers within the same autonomous system can exchange routing information.

```
D        10.10.0.0/16 [90/2297856] via 172.20.12.1, 00:21:23, Serial1/0
D EX     10.10.1.0/24 [170/2195456] via 172.20.12.1, 00:21:23, Serial1/0
D EX     10.10.2.0/24 [170/2195456] via 172.20.12.1, 00:21:23, Serial1/0
C        10.20.41.0/24 is directly connected, Loopback2
L        10.20.41.1/32 is directly connected, Loopback2
C        10.20.89.0/24 is directly connected, Loopback3
L        10.20.89.1/32 is directly connected, Loopback3
C        10.20.97.0/24 is directly connected, Loopback1
L        10.20.97.1/32 is directly connected, Loopback1
D EX     10.30.16.0/20 [170/2195456] via 172.20.12.1, 00:20:30, Serial1/0
D        10.100.1.0/24 [90/2297856] via 172.20.12.1, 00:21:23, Serial1/0
      100.0.0.0/30 is subnetted, 1 subnets
D        100.100.102.0 [90/27392000] via 172.20.12.1, 00:21:16, Serial1/0
      172.20.0.0/16 is variably subnetted, 5 subnets, 4 masks
C        172.20.2.0/24 is directly connected, FastEthernet0/0
L        172.20.2.1/32 is directly connected, FastEthernet0/0
D        172.20.11.0/28 [90/2681856] via 172.20.12.1, 00:21:23, Serial1/0
C        172.20.12.0/30 is directly connected, Serial1/0
L        172.20.12.2/32 is directly connected, Serial1/0
      172.21.0.0/16 is variably subnetted, 2 subnets, 2 masks
D        172.21.11.0/28 [90/2681856] via 172.20.12.1, 00:21:23, Serial1/0
D EX     172.21.12.0/30 [170/2195456] via 172.20.12.1, 00:21:20, Serial1/0
```

The `show ip route` output shows that EIGRP-learned routes (denoted with "D") have been successfully added to the routing table, confirming proper configuration and full connectivity within the EIGRP domain.

#3: OSPF Configuration - #`show ip ospf neighbour & show ip route`
Output/Result: (You may snapshot from the simulation-GNS3 or PT).

ER1 [2m] - `show ip ospf neighbor`

```
ER1#sh ip ospf neigh

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:31    172.21.11.1     Serial5/0
```

OR1 [2m] – `show ip route`

```
      10.0.0.0/8 is variably subnetted, 13 subnets, 4 masks
O        10.2.2.1/32 [110/2] via 172.21.12.2, 00:13:18, FastEthernet0/0
O        10.2.20.0/24 [110/2] via 172.21.12.2, 00:13:18, FastEthernet0/0
O IA     10.3.3.1/32 [110/66] via 172.21.12.2, 00:13:18, FastEthernet0/0
O E2     10.10.0.0/16 [110/20] via 172.21.11.11, 00:13:52, Serial1/0
O E2     10.10.1.0/24 [110/20] via 172.21.11.11, 00:13:52, Serial1/0
O E2     10.10.2.0/24 [110/20] via 172.21.11.11, 00:13:52, Serial1/0
O E1     10.20.41.0/24 [110/84] via 172.21.11.11, 00:13:52, Serial1/0
O E1     10.20.89.0/24 [110/84] via 172.21.11.11, 00:13:52, Serial1/0
O E1     10.20.97.0/24 [110/84] via 172.21.11.11, 00:13:52, Serial1/0
O IA     10.30.16.0/20 [110/66] via 172.21.12.2, 00:13:18, FastEthernet0/0
O E2     10.100.1.0/24 [110/20] via 172.21.11.11, 00:13:52, Serial1/0
      100.0.0.0/30 is subnetted, 1 subnets
O E2     100.100.102.0 [110/20] via 172.21.11.11, 00:13:52, Serial1/0
      172.20.0.0/16 is variably subnetted, 3 subnets, 3 masks
O E1     172.20.2.0/24 [110/84] via 172.21.11.11, 00:13:52, Serial1/0
O E2     172.20.11.0/28 [110/20] via 172.21.11.11, 00:13:52, Serial1/0
O E2     172.20.12.0/30 [110/20] via 172.21.11.11, 00:13:52, Serial1/0
      172.23.0.0/29 is subnetted, 1 subnets
O IA     172.23.23.0 [110/65] via 172.21.12.2, 00:13:18, FastEthernet0/0
      192.168.2.0/32 is subnetted, 1 subnets
O        192.168.2.1 [110/2] via 172.21.12.2, 00:13:18, FastEthernet0/0
      192.168.3.0/32 is subnetted, 1 subnets
O        192.168.3.1 [110/2] via 172.21.12.2, 00:13:18, FastEthernet0/0
      192.168.4.0/32 is subnetted, 1 subnets
O        192.168.4.1 [110/2] via 172.21.12.2, 00:13:18, FastEthernet0/0
```

OR2 [2m] - `show ip ospf neighbor`

```
OR2#sh ip ospf neigh

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:30    172.21.12.1     FastEthernet0/0
3.3.3.3           0   FULL/  -           -        172.23.23.3     OSPF_VL0
3.3.3.3           0   FULL/  -        00:00:35    172.23.23.3     Serial1/0
```

OR3 [2m] – `show ip route`

```
O*E1  0.0.0.0/0 [110/66] via 172.23.23.2, 00:11:19, Serial1/0
      10.0.0.0/8 is variably subnetted, 19 subnets, 4 masks
O        10.1.1.1/32 [110/66] via 172.23.23.2, 00:11:19, Serial1/0
O        10.2.2.1/32 [110/65] via 172.23.23.2, 00:11:55, Serial1/0
O        10.2.20.0/24 [110/65] via 172.23.23.2, 00:11:55, Serial1/0
O E2     10.10.0.0/16 [110/20] via 172.23.23.2, 00:11:19, Serial1/0
O E2     10.10.1.0/24 [110/20] via 172.23.23.2, 00:11:19, Serial1/0
O E2     10.10.2.0/24 [110/20] via 172.23.23.2, 00:11:19, Serial1/0
O E1     10.20.41.0/24 [110/149] via 172.23.23.2, 00:11:19, Serial1/0
O E1     10.20.89.0/24 [110/149] via 172.23.23.2, 00:11:19, Serial1/0
O E1     10.20.97.0/24 [110/149] via 172.23.23.2, 00:11:19, Serial1/0
O        10.30.16.0/20 is a summary, 00:12:07, Null0
O E2     10.100.1.0/24 [110/20] via 172.23.23.2, 00:11:19, Serial1/0
      100.0.0.0/30 is subnetted, 1 subnets
O E2     100.100.102.0 [110/20] via 172.23.23.2, 00:11:19, Serial1/0
      172.20.0.0/16 is variably subnetted, 3 subnets, 3 masks
O E1     172.20.2.0/24 [110/149] via 172.23.23.2, 00:11:19, Serial1/0
O E2     172.20.11.0/28 [110/20] via 172.23.23.2, 00:11:19, Serial1/0
O E2     172.20.12.0/30 [110/20] via 172.23.23.2, 00:11:19, Serial1/0
      172.21.0.0/16 is variably subnetted, 2 subnets, 2 masks
O        172.21.11.0/28 [110/129] via 172.23.23.2, 00:11:19, Serial1/0
O        172.21.12.0/30 [110/65] via 172.23.23.2, 00:11:55, Serial1/0
      192.168.2.0/32 is subnetted, 1 subnets
O        192.168.2.1 [110/65] via 172.23.23.2, 00:11:55, Serial1/0
      192.168.3.0/32 is subnetted, 1 subnets
O        192.168.3.1 [110/65] via 172.23.23.2, 00:11:55, Serial1/0
      192.168.4.0/32 is subnetted, 1 subnets
O        192.168.4.1 [110/65] via 172.23.23.2, 00:11:55, Serial1/0
      201.122.12.0/30 is subnetted, 1 subnets
O        201.122.12.0 [110/129] via 172.23.23.2, 00:11:19, Serial1/0
```

Explanation:

The `show ip ospf neighbor` output shows that OSPF adjacencies have been successfully formed between routers, meaning they can exchange routing information. The `show ip route` confirms that OSPF-learned routes are present in the routing table, proving that routing updates are working and connectivity is established across OSPF routers.

#4: Network Routes updated from OSPF area 30 - #`show ip ospf neighbor & show ip route`
Output: (You may snapshot from the simulation-GNS3 or PT)

OR1 [2m] - `show ip route`

```
O E2     10.10.2.0/24 [110/20] via 172.21.11.11, 00:37:51, Serial1/0
O E1     10.20.41.0/24 [110/84] via 172.21.11.11, 00:37:51, Serial1/0
O E1     10.20.89.0/24 [110/84] via 172.21.11.11, 00:37:51, Serial1/0
O E1     10.20.97.0/24 [110/84] via 172.21.11.11, 00:37:51, Serial1/0
O IA     10.30.16.1/32 [110/66] via 172.21.12.2, 00:00:08, FastEthernet0/0
O IA     10.30.17.1/32 [110/66] via 172.21.12.2, 00:00:08, FastEthernet0/0
O IA     10.30.18.1/32 [110/66] via 172.21.12.2, 00:00:08, FastEthernet0/0
O E2     10.100.1.0/24 [110/20] via 172.21.11.11, 00:37:51, Serial1/0
      100.0.0.0/30 is subnetted, 1 subnets
O E2     100.100.102.0 [110/20] via 172.21.11.11, 00:37:51, Serial1/0
      172.20.0.0/16 is variably subnetted, 3 subnets, 3 masks
O E1     172.20.2.0/24 [110/84] via 172.21.11.11, 00:37:51, Serial1/0
```

OR2 [2m] - `show ip ospf neighbour & show ip route`

```
OR2#sh ip ospf neigh

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:38    172.21.12.1     FastEthernet0/0
3.3.3.3           0   FULL/  -           -        172.23.23.3     OSPF_VL0
3.3.3.3           0   FULL/  -        00:00:30    172.23.23.3     Serial1/0
```

```
O E2     10.10.2.0/24 [110/20] via 172.21.11.11, 00:37:51, Serial1/0
O E1     10.20.41.0/24 [110/84] via 172.21.11.11, 00:37:51, Serial1/0
O E1     10.20.89.0/24 [110/84] via 172.21.11.11, 00:37:51, Serial1/0
O E1     10.20.97.0/24 [110/84] via 172.21.11.11, 00:37:51, Serial1/0
O IA     10.30.16.1/32 [110/66] via 172.21.12.2, 00:00:08, FastEthernet0/0
O IA     10.30.17.1/32 [110/66] via 172.21.12.2, 00:00:08, FastEthernet0/0
O IA     10.30.18.1/32 [110/66] via 172.21.12.2, 00:00:08, FastEthernet0/0
O E2     10.100.1.0/24 [110/20] via 172.21.11.11, 00:37:51, Serial1/0
      100.0.0.0/30 is subnetted, 1 subnets
O E2     100.100.102.0 [110/20] via 172.21.11.11, 00:37:51, Serial1/0
      172.20.0.0/16 is variably subnetted, 3 subnets, 3 masks
O E1     172.20.2.0/24 [110/84] via 172.21.11.11, 00:37:51, Serial1/0
```

OR3 Routing Table [2m] - `show ip ospf neighbour`

```
OR3# sh ip ospf neigh

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           0   FULL/  -           -        172.23.23.2     OSPF_VL0
2.2.2.2           0   FULL/  -        00:00:30    172.23.23.2     Serial1/0
```

Explanation [2m]

The show ip route prompt is OSPF Inter Area that the 10.20.17.1/22 and 10.20.18.1/32 are both O IA routes. The sh ip ospf neighbor will show the all neighbor id include the address and the interface.

#5: eBGP Configuration - #`show ip bgp sum` & `show ip bgp`

Output/Result: (You may snapshot from the simulation-GNS3 or PT).

HQ [2m] - `show ip bgp & show ip bgp sum`

```
HQ#sh ip bgp
BGP table version is 12, local router ID is 210.11.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  100.100.102.0/30 210.200.20.1             0             0 200 i
 *>  172.20.11.0/28   0.0.0.0                  0         32768 ?
 *>  200.100.20.0     210.200.20.1             0             0 200 i
 *>  201.110.11.0/28  0.0.0.0                  0         32768 i
 *   201.120.12.0/30  210.200.20.1             0             0 200 ?
 *>                   0.0.0.0                  0         32768 ?
 *>  201.122.12.0/30  210.200.20.1             0             0 200 ?
 *>  201.223.23.0/29  210.200.20.1             0             0 200 ?
 *>  201.223.32.0/29  210.200.20.1             0             0 200 i
 *>  210.11.1.0       0.0.0.0                  0         32768 i
 *>  210.200.20.0     210.200.20.1             0             0 200 i
 *>  220.202.20.0     210.200.20.1             0             0 200 i
```

```
HQ# sh ip bgp sum
BGP router identifier 210.11.1.1, local AS number 65100
BGP table version is 93, main routing table version 93
40 network entries using 5760 bytes of memory
48 path entries using 3840 bytes of memory
24/20 BGP path/bestpath attribute entries using 3456 bytes of memory
6 BGP AS-PATH entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 13200 total bytes of memory
BGP activity 40/0 prefixes, 363/315 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.100.1.1      4        65010      35      50       93    0    0 00:19:13       23
210.11.11.1     4          100     193      49       93    0    0 00:19:55       10
210.200.20.1    4          200      23      49       93    0    0 00:19:52       11
```

BG3 [2m] - `show ip bgp & show ip bgp sum`

```
     Network          Next Hop            Metric LocPrf Weight Path
 *   100.100.102.0/30 192.168.20.1                           0 100 ?
 *>                   220.202.20.1             0    200      0 200 i
 *   172.20.11.0/28   192.168.20.1                           0 100 ?
 *>                   220.202.20.1                  200      0 200 65100 ?
 *>  192.121.11.0/28  220.202.20.1                  200      0 200 65100 100 ?
 *                    192.168.20.1         30720             0 100 ?
 *>  192.122.12.0/30  220.202.20.1                  200      0 200 65100 100 ?
 *                    192.168.20.1             0             0 100 ?
 *>  192.168.10.0     220.202.20.1                  200      0 200 65100 100 ?
 *                    192.168.20.1        284160             0 100 ?
 *>  192.168.11.0     220.202.20.1                  200      0 200 65100 100 i
 *                    192.168.20.1        158720             0 100 ?
 *>  192.168.20.0     192.168.20.1             0             0 100 i
 *>  192.168.30.0     0.0.0.0                  0         32768 i
     Network          Next Hop            Metric LocPrf Weight Path
 *   200.100.20.0     192.168.20.1                           0 100 ?
 *>                   220.202.20.1             0    200      0 200 i
 *   201.110.11.0/28  192.168.20.1       2174976             0 100 ?
 *>                   220.202.20.1                  200      0 200 65100 i
 *   201.120.12.0/30  192.168.20.1                           0 100 ?
 *>                   220.202.20.1             0    200      0 200 ?
 *   201.122.12.0/30  192.168.20.1                           0 100 ?
 *>                   220.202.20.1             0    200      0 200 ?
 r   201.223.23.0/29  192.168.20.1                           0 100 ?
 r>                   220.202.20.1             0    200      0 200 ?
 r   201.223.32.0/29  192.168.20.1                           0 100 ?
 r>                   220.202.20.1             0    200      0 200 i
 r>  202.230.23.0/29  220.202.20.1                  200      0 200 65100 100 ?
 r                    192.168.20.1             0             0 100 ?
 *>  203.23.30.0      0.0.0.0                  0         32768 i
 *   210.11.1.0       192.168.20.1                           0 100 65100 i
 *>                   220.202.20.1                  200      0 200 65100 i
 *>  210.11.11.0      220.202.20.1                  200      0 200 65100 100 i
 *                    192.168.20.1        158720             0 100 ?
 *   210.200.20.0     192.168.20.1                           0 100 ?
 *>                   220.202.20.1             0    200      0 200 i
 *   220.202.20.0     192.168.20.1                           0 100 ?
     Network          Next Hop            Metric LocPrf Weight Path
 *>                   220.202.20.1             0    200      0 200 i
```

```
BG3#sh ip bgp sum
BGP router identifier 192.168.30.1, local AS number 300
BGP table version is 954, main routing table version 954
40 network entries using 5760 bytes of memory
40 path entries using 3200 bytes of memory
10/10 BGP path/bestpath attribute entries using 1440 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 10448 total bytes of memory
BGP activity 40/0 prefixes, 97/57 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
192.168.20.1    4          100     185      70      954    0    0 00:16:05       38
220.202.20.1    4          200      20      71      954    0    0 00:16:05        0
```

ISP2 [2m] `show ip bgp & show ip bgp sum`

Explanation [2m]

1. show ip bgp

Displays the BGP routing table, showing:

Best paths (>) selected for forwarding

Next-hop IPs, AS_PATHs, and routing attributes (Weight, Local Pref, MED)

```
ISP2#sh ip bgp
BGP table version is 22, local router ID is 220.202.20.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  100.100.102.0/30 0.0.0.0                  0         32768 i
 *>  172.20.11.0/28   210.11.1.1               0             0 65100 ?
 *>  192.121.11.0/28  210.11.1.1                             0 65100 100 ?
 *>  192.122.12.0/30  210.11.1.1                             0 65100 100 ?
 *>  192.168.10.0     210.11.1.1                             0 65100 100 ?
 *>  192.168.11.0     210.11.1.1                             0 65100 100 i
 *>  192.168.20.0     192.168.30.1                           0 300 100 i
 *                    210.11.1.1                             0 65100 100 ?
 *   192.168.30.0     210.11.1.1                             0 65100 100 ?
 *>                   192.168.30.1             0             0 300 i
 *>  200.100.20.0     0.0.0.0                  0         32768 i
 *>  201.110.11.0/28  210.11.1.1               0             0 65100 i
 *>  201.120.12.0/30  0.0.0.0                  0         32768 ?
 *                    210.11.1.1               0             0 65100 ?
     Network          Next Hop            Metric LocPrf Weight Path
 *>  201.122.12.0/30  0.0.0.0                  0         32768 ?
 *>  201.223.23.0/29  0.0.0.0                  0         32768 ?
 *>  201.223.32.0/29  0.0.0.0                  0         32768 i
 *>  202.230.23.0/29  210.11.1.1                             0 65100 100 ?
 *   203.23.30.0      210.11.1.1                             0 65100 100 ?
 *>                   192.168.30.1             0             0 300 i
 *>  210.11.1.0       210.11.1.1               0             0 65100 i
 *>  210.11.11.0      210.11.1.1                             0 65100 100 i
 *>  210.200.20.0     0.0.0.0                  0         32768 i
 *>  220.202.20.0     0.0.0.0                  0         32768 i
```

```
ISP2#sh ip bgp sum
BGP router identifier 220.202.20.1, local AS number 200
BGP table version is 42, main routing table version 42
40 network entries using 5760 bytes of memory
82 path entries using 6560 bytes of memory
12/9 BGP path/bestpath attribute entries using 1728 bytes of memory
7 BGP AS-PATH entries using 168 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 14216 total bytes of memory
BGP activity 40/0 prefixes, 134/52 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
192.168.30.1    4          300      72      21       42    0    0 00:16:25       40
210.11.1.1      4        65100      51      24       42    0    0 00:16:21       34
```

Example: 210.11.1.0 via AS 65100 (ISP2) or locally originated (HQ).

2. show ip bgp summary

Shows BGP neighbor status:

Peering sessions (IP, AS, Up/Down time)

Memory usage and message counts (MsgRcvd/MsgSent)

Example: HQ peers with AS 65000 (10.100.1.1) for ~60 minutes.

Summary:

show ip bgp = Routes & path selection

show ip bgp sum = Neighbor health & session stats.

#6: iBGP Configuration. - # show ip bgp sum & show ip bgp

Screenshot the output then highlight the particular output and explain it.

ISP1 - show ip bgp sum & - show ip bgp

```
ISP1#sh ip bgp sum
BGP router identifier 210.11.11.1, local AS number 100
BGP table version is 479924, main routing table version 479924
40 network entries using 5760 bytes of memory
73 path entries using 5840 bytes of memory
22/13 BGP path/bestpath attribute entries using 3168 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 14864 total bytes of memory
BGP activity 40/0 prefixes, 245381/245308 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
192.168.10.1    4          100   66683   77090   479924    0    0 00:37:08        5
192.168.20.1    4          100   44309   77090   479924    0    0 00:37:08       29
210.11.1.1      4        65100      64     257   475855    0    0 00:37:06       31
```

BG1 - show ip bgp sum & - show ip bgp

```
BG1#sh ip bgp sum
BGP router identifier 192.168.10.1, local AS number 100
BGP table version is 808860, main routing table version 808860
40 network entries using 5760 bytes of memory
76 path entries using 6080 bytes of memory
13/6 BGP path/bestpath attribute entries using 1872 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 13808 total bytes of memory
BGP activity 58/18 prefixes, 443151/443075 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
192.168.11.1    4          100   79037   68381   808860    0    0 00:38:07       34
192.168.20.1    4          100   45434   68381   808860    0    0 00:38:05        3
```

```
     Network          Next Hop            Metric LocPrf Weight Path
 *>  100.100.102.0/30 192.121.11.11       284160         32768 ?
 *>  172.20.11.0/28   192.121.11.11       284160         32768 ?
 * i 192.121.11.0/28  192.168.11.1             0    100      0 ?
 *>                   0.0.0.0                  0         32768 ?
 * i 192.122.12.0/30  192.168.20.1             0    100      0 ?
 *>                   0.0.0.0                  0         32768 ?
 *>  192.168.10.0     0.0.0.0                  0         32768 i
 * i 192.168.11.0     192.168.11.1             0    100      0 i
 *>                   192.121.11.11       156160         32768 ?
 * i 192.168.20.0     192.168.20.1             0    100      0 i
 *>                   192.122.12.2        156160         32768 ?
 *>i 192.168.30.0     192.168.30.1             0    100      0 300 i
 *>  200.100.20.0     192.121.11.11       284160         32768 ?
 * i 201.110.11.0/28  192.168.11.1             0    100      0 ?
     Network          Next Hop            Metric LocPrf Weight Path
 *>                   192.121.11.11      2172416         32768 ?
 *>  201.120.12.0/30  192.121.11.11       284160         32768 ?
 * i                  210.11.1.1               0    100      0 65100 ?
 *>  201.122.12.0/30  192.121.11.11       284160         32768 ?
 *>  201.223.23.0/29  192.121.11.11       284160         32768 ?
 *>  201.223.32.0/29  192.121.11.11       284160         32768 ?
 * i 202.230.23.0/29  192.168.20.1             0    100      0 ?
 *>                   192.122.12.2       2172416         32768 ?
 *>  203.23.30.0      192.122.12.2        284160         32768 ?
 r>i 210.11.1.0       210.11.1.1               0    100      0 65100 i
 * i 210.11.11.0      192.168.11.1             0    100      0 i
 *>                   192.121.11.11       156160         32768 ?
 *>  210.200.20.0     192.121.11.11       284160         32768 ?
 *>  220.202.20.0     192.121.11.11       284160         32768 ?
```

```
     Network          Next Hop            Metric LocPrf Weight Path
*>i 100.100.102.0/30 192.122.12.1        286720    100      0 ?
*                    210.11.1.1                             0 65100 200 i
*>i 172.20.11.0/28   192.122.12.1        286720    100      0 ?
*                    210.11.1.1               0             0 65100 ?
* i 192.121.11.0/28  192.122.12.1         30720    100      0 ?
* i                  192.168.10.1             0    100      0 ?
*>                   0.0.0.0                  0         32768 ?
* i 192.122.12.0/30  192.168.20.1             0    100      0 ?
* i                  192.168.10.1             0    100      0 ?
*>                   192.121.11.1         30720         32768 ?
* i 192.168.10.0     192.122.12.1        284160    100      0 ?
* i                  192.168.10.1             0    100      0 i
*>                   192.121.11.1        284160         32768 ?
* i 192.168.11.0     192.122.12.1        158720    100      0 ?
     Network          Next Hop            Metric LocPrf Weight Path
*>                   0.0.0.0                  0         32768 i
* i 192.168.20.0     192.168.20.1             0    100      0 i
* i                  192.122.12.2        156160    100      0 ?
*>                   192.121.11.1        158720         32768 ?
*>i 192.168.30.0     192.168.30.1             0    100      0 300 i
*>i 200.100.20.0     192.122.12.1        286720    100      0 ?
*                    210.11.1.1                             0 65100 200 i
* i 201.110.11.0/28  192.122.12.1       2174976    100      0 ?
*                    210.11.1.1               0             0 65100 i
*>                   0.0.0.0                  0         32768 ?
*>i 201.120.12.0/30  192.122.12.1        286720    100      0 ?
*                    210.11.1.1               0             0 65100 ?
*>i 201.122.12.0/30  192.122.12.1        286720    100      0 ?
*                    210.11.1.1                             0 65100 200 ?
*>i 201.223.23.0/29  192.122.12.1        286720    100      0 ?
*                    210.11.1.1                             0 65100 200 ?
*>i 201.223.32.0/29  192.122.12.1        286720    100      0 ?
*                    210.11.1.1                             0 65100 200 i
* i 202.230.23.0/29  192.168.20.1             0    100      0 ?
* i                  192.122.12.2       2172416    100      0 ?
*>                   192.121.11.1       2174976         32768 ?
*>i 203.23.30.0      192.168.30.1             0    100      0 300 i
     Network          Next Hop            Metric LocPrf Weight Path
*>  210.11.1.0       210.11.1.1               0             0 65100 i
* i 210.11.11.0      192.122.12.1        158720    100      0 ?
*>                   0.0.0.0                  0         32768 i
*>i 210.200.20.0     192.122.12.1        286720    100      0 ?
*                    210.11.1.1                             0 65100 200 i
*>i 220.202.20.0     192.122.12.1        286720    100      0 ?
*                    210.11.1.1                             0 65100 200 i
```

BG2 - `show ip bgp sum` & - `show ip bgp`

```
BG2#sh ip bgp sum
BGP router identifier 192.168.20.1, local AS number 100
BGP table version is 1095648, main routing table version 1095648
40 network entries using 5760 bytes of memory
56 path entries using 4480 bytes of memory
22/12 BGP path/bestpath attribute entries using 3168 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 13504 total bytes of memory
BGP activity 40/0 prefixes, 639827/639771 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
192.168.10.1    4          100   68989   45834  1095648    0    0 00:38:22       25
192.168.11.1    4          100   79754   45834  1095648    0    0 00:38:25       21
192.168.30.1    4          300     121     278  1093841    0    0 00:38:23        2
```

```
     Network          Next Hop            Metric LocPrf Weight Path
* i 100.100.102.0/30 210.11.1.1               0    100      0 65100 200 i
*>                   192.168.30.1                           0 300 200 i
*>  172.20.11.0/28   192.122.12.1        286720         32768 ?
* i                  192.121.11.11       284160    100      0 ?
* i                  210.11.1.1               0    100      0 65100 ?
*                    192.168.30.1                           0 300 200 65100 ?
* i 192.121.11.0/28  192.168.11.1             0    100      0 ?
* i                  192.168.10.1             0    100      0 ?
*>                   192.122.12.1         30720         32768 ?
* i 192.122.12.0/30  192.121.11.1         30720    100      0 ?
* i                  192.168.10.1             0    100      0 ?
*>                   0.0.0.0                  0         32768 ?
* i 192.168.10.0     192.121.11.1        284160    100      0 ?
* i                  192.168.10.1             0    100      0 i
     Network          Next Hop            Metric LocPrf Weight Path
*>                   192.122.12.1        284160         32768 ?
* i 192.168.11.0     192.168.11.1             0    100      0 i
* i                  192.121.11.11       156160    100      0 ?
*>                   192.122.12.1        158720         32768 ?
* i 192.168.20.0     192.121.11.1        158720    100      0 ?
*>                   0.0.0.0                  0         32768 i
*>  192.168.30.0     192.168.30.1             0             0 300 i
* i 200.100.20.0     210.11.1.1               0    100      0 65100 200 i
*>                   192.168.30.1                           0 300 200 i
* i 201.110.11.0/28  192.168.11.1             0    100      0 ?
*                    192.168.30.1                           0 300 200 65100 i
* i                  192.121.11.11      2172416    100      0 ?
*>                   192.122.12.1       2174976         32768 ?
*>  201.120.12.0/30  192.122.12.1        286720         32768 ?
* i                  192.121.11.11       284160    100      0 ?
* i                  210.11.1.1               0    100      0 65100 ?
*                    192.168.30.1                           0 300 200 ?
* i 201.122.12.0/30  210.11.1.1               0    100      0 65100 200 ?
*>                   192.168.30.1                           0 300 200 ?
* i 201.223.23.0/29  210.11.1.1               0    100      0 65100 200 ?
*>                   192.168.30.1                           0 300 200 ?
* i 201.223.32.0/29  210.11.1.1               0    100      0 65100 200 i
```

Explanation

`show ip bgp`

Displays the BGP routing table, showing:

- Best path (>) marked for forwarding
- Next-hop IPs, AS_PATH, and route type (i – iBGP)
- Example: Route learned from iBGP neighbor 192.168.10.1 with internal AS path and valid next-hop.

`show ip bgp summary`

Shows iBGP neighbor status:

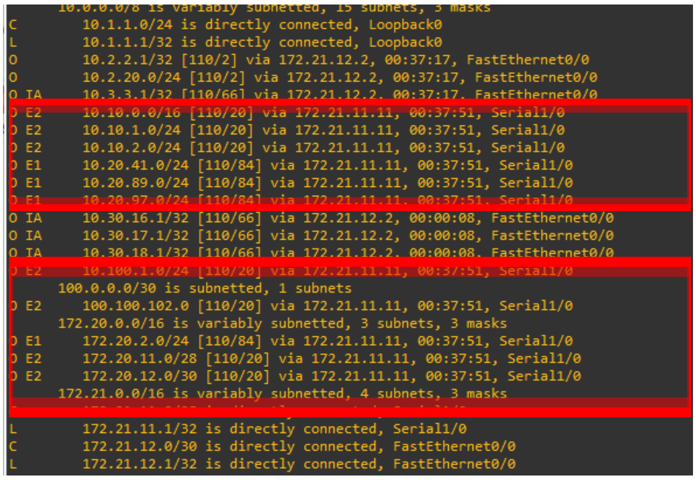
- Peer IPs, AS numbers, uptime, and prefix received
- MsgRcvd/MsgSent confirm active sessions
- Example: ISP1 has iBGP peers 192.168.10.1 (BG1) and 192.168.20.1 (BG2), all in the established state, with prefixes received.

#7: Routing update from different routing protocols. - `# show ip route`

Screenshot the output then highlight the particular output and explain it.

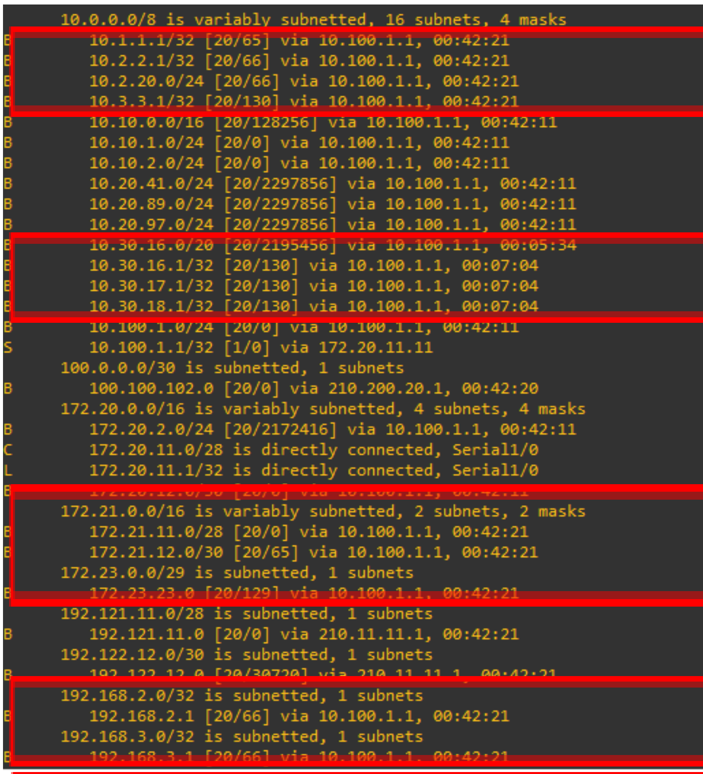
Output/Result [OSPF – EIGRP]: [6m]

OR1 [2m] – highlight network routes from EIGRP

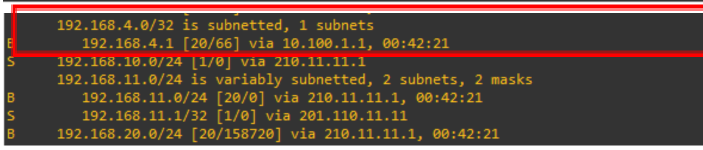


```
      10.0.0.0/8 is variably subnetted, 15 subnets, 3 masks
C        10.1.1.0/24 is directly connected, Loopback0
L        10.1.1.1/32 is directly connected, Loopback0
O        10.2.2.1/32 [110/2] via 172.21.12.2, 00:37:17, FastEthernet0/0
O        10.2.20.0/24 [110/2] via 172.21.12.2, 00:37:17, FastEthernet0/0
O IA     10.3.3.1/32 [110/66] via 172.21.12.2, 00:37:17, FastEthernet0/0
O E2     10.10.0.0/16 [110/20] via 172.21.11.11, 00:37:51, Serial1/0
O E2     10.10.1.0/24 [110/20] via 172.21.11.11, 00:37:51, Serial1/0
O E2     10.10.2.0/24 [110/20] via 172.21.11.11, 00:37:51, Serial1/0
O E1     10.20.41.0/24 [110/84] via 172.21.11.11, 00:37:51, Serial1/0
O E1     10.20.89.0/24 [110/84] via 172.21.11.11, 00:37:51, Serial1/0
O E1     10.20.97.0/24 [110/84] via 172.21.11.11, 00:37:51, Serial1/0
O IA     10.30.16.1/32 [110/66] via 172.21.12.2, 00:00:08, FastEthernet0/0
O IA     10.30.17.1/32 [110/66] via 172.21.12.2, 00:00:08, FastEthernet0/0
O IA     10.30.18.1/32 [110/66] via 172.21.12.2, 00:00:08, FastEthernet0/0
O E2     10.100.1.0/24 [110/20] via 172.21.11.11, 00:37:51, Serial1/0
      100.0.0.0/30 is subnetted, 1 subnets
O E2     100.100.102.0 [110/20] via 172.21.11.11, 00:37:51, Serial1/0
      172.20.0.0/16 is variably subnetted, 3 subnets, 3 masks
O E1     172.20.2.0/24 [110/84] via 172.21.11.11, 00:37:51, Serial1/0
O E2     172.20.11.0/28 [110/20] via 172.21.11.11, 00:37:51, Serial1/0
O E2     172.20.12.0/30 [110/20] via 172.21.11.11, 00:37:51, Serial1/0
      172.21.0.0/16 is variably subnetted, 4 subnets, 3 masks
L        172.21.11.1/32 is directly connected, Serial1/0
C        172.21.12.0/30 is directly connected, FastEthernet0/0
L        172.21.12.1/32 is directly connected, FastEthernet0/0
```

HQ – highlight network routes from OSPF



```
      10.0.0.0/8 is variably subnetted, 16 subnets, 4 masks
B        10.1.1.1/32 [20/65] via 10.100.1.1, 00:42:21
B        10.2.2.1/32 [20/66] via 10.100.1.1, 00:42:21
B        10.2.20.0/24 [20/66] via 10.100.1.1, 00:42:21
B        10.3.3.1/32 [20/130] via 10.100.1.1, 00:42:21
B        10.10.0.0/16 [20/128256] via 10.100.1.1, 00:42:11
B        10.10.1.0/24 [20/0] via 10.100.1.1, 00:42:11
B        10.10.2.0/24 [20/0] via 10.100.1.1, 00:42:11
B        10.20.41.0/24 [20/2297856] via 10.100.1.1, 00:42:11
B        10.20.89.0/24 [20/2297856] via 10.100.1.1, 00:42:11
B        10.20.97.0/24 [20/2297856] via 10.100.1.1, 00:42:11
B        10.30.16.0/20 [20/2195456] via 10.100.1.1, 00:05:34
B        10.30.16.1/32 [20/130] via 10.100.1.1, 00:07:04
B        10.30.17.1/32 [20/130] via 10.100.1.1, 00:07:04
B        10.30.18.1/32 [20/130] via 10.100.1.1, 00:07:04
B        10.100.1.0/24 [20/0] via 10.100.1.1, 00:42:11
S        10.100.1.1/32 [1/0] via 172.20.11.11
      100.0.0.0/30 is subnetted, 1 subnets
B        100.100.102.0 [20/0] via 210.200.20.1, 00:42:20
      172.20.0.0/16 is variably subnetted, 4 subnets, 4 masks
B        172.20.2.0/24 [20/2172416] via 10.100.1.1, 00:42:11
C        172.20.11.0/28 is directly connected, Serial1/0
L        172.20.11.1/32 is directly connected, Serial1/0
B        172.20.12.0/30 [20/0] via 10.100.1.1, 00:42:11
      172.21.0.0/16 is variably subnetted, 2 subnets, 2 masks
B        172.21.11.0/28 [20/0] via 10.100.1.1, 00:42:21
B        172.21.12.0/30 [20/65] via 10.100.1.1, 00:42:21
      172.23.0.0/29 is subnetted, 1 subnets
B        172.23.23.0 [20/129] via 10.100.1.1, 00:42:21
      192.121.11.0/28 is subnetted, 1 subnets
B        192.121.11.0 [20/0] via 210.11.11.1, 00:42:21
      192.122.12.0/30 is subnetted, 1 subnets
B        192.122.12.0 [20/30720] via 210.11.11.1, 00:42:21
      192.168.2.0/32 is subnetted, 1 subnets
B        192.168.2.1 [20/66] via 10.100.1.1, 00:42:21
      192.168.3.0/32 is subnetted, 1 subnets
B        192.168.3.1 [20/66] via 10.100.1.1, 00:42:21
```



```
      192.168.4.0/32 is subnetted, 1 subnets
B        192.168.4.1 [20/66] via 10.100.1.1, 00:42:21
S     192.168.10.0/24 [1/0] via 210.11.11.1
      192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
B        192.168.11.0/24 [20/0] via 210.11.11.1, 00:42:21
S        192.168.11.1/32 [1/0] via 201.110.11.11
B     192.168.20.0/24 [20/158720] via 210.11.11.1, 00:42:21
```

Explanation

OR1

The routing table shows EIGRP-learned routes for networks like 10.20.41.0/24, 10.20.49.0/24

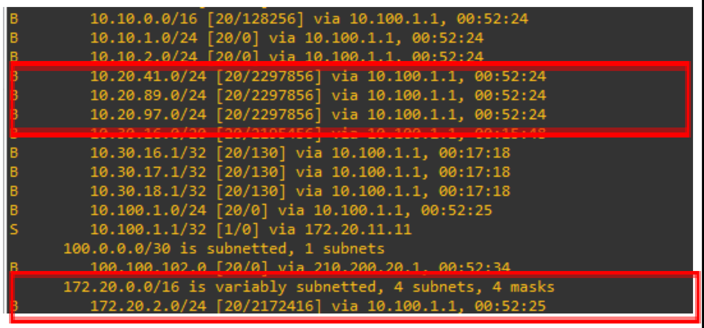
These routes are learned via EIGRP AS 100 through Serial1/0, confirming successful EIGRP neighbor formation and route exchange.

HQ

The routing table displays BGP learned routes, such as 10.0.0.0/8, 172.21.0.0/16 and 192.168.2.0/24

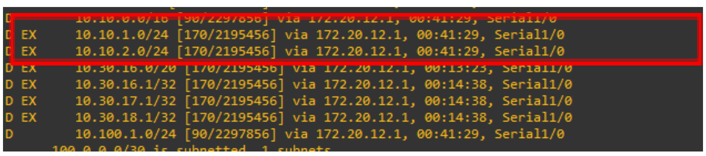
These routes are received from BGP peer 10.100.1.1, verifying that external BGP (eBGP) is active and routing updates are working correctly.

HQ - [2m] – highlight network routes from EIGRP AS200

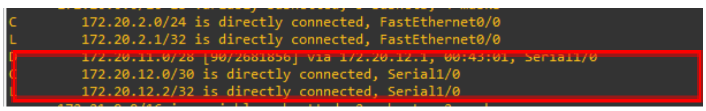


```
B        10.10.0.0/16 [20/128256] via 10.100.1.1, 00:52:24
B        10.10.1.0/24 [20/0] via 10.100.1.1, 00:52:24
B        10.10.2.0/24 [20/0] via 10.100.1.1, 00:52:24
B        10.20.41.0/24 [20/2297856] via 10.100.1.1, 00:52:24
B        10.20.89.0/24 [20/2297856] via 10.100.1.1, 00:52:24
B        10.20.97.0/24 [20/2297856] via 10.100.1.1, 00:52:24
B        10.30.16.0/20 [20/2195456] via 10.100.1.1, 00:15:48
B        10.30.16.1/32 [20/130] via 10.100.1.1, 00:17:18
B        10.30.17.1/32 [20/130] via 10.100.1.1, 00:17:18
B        10.30.18.1/32 [20/130] via 10.100.1.1, 00:17:18
B        10.100.1.0/24 [20/0] via 10.100.1.1, 00:52:25
S        10.100.1.1/32 [1/0] via 172.20.11.11
      100.0.0.0/30 is subnetted, 1 subnets
B        100.100.102.0 [20/0] via 210.200.20.1, 00:52:34
      172.20.0.0/16 is variably subnetted, 4 subnets, 4 masks
B        172.20.2.0/24 [20/2172416] via 10.100.1.1, 00:52:25
```

ER2 - [2m] – highlight network routes from EIGRP AS100



```
D        10.10.0.0/16 [90/2297856] via 172.20.12.1, 00:41:29, Serial1/0
D EX     10.10.1.0/24 [170/2195456] via 172.20.12.1, 00:41:29, Serial1/0
D EX     10.10.2.0/24 [170/2195456] via 172.20.12.1, 00:41:29, Serial1/0
D EX     10.30.16.0/20 [170/2195456] via 172.20.12.1, 00:13:23, Serial1/0
D EX     10.30.16.1/32 [170/2195456] via 172.20.12.1, 00:14:38, Serial1/0
D EX     10.30.17.1/32 [170/2195456] via 172.20.12.1, 00:14:38, Serial1/0
D EX     10.30.18.1/32 [170/2195456] via 172.20.12.1, 00:14:38, Serial1/0
D        10.100.1.0/24 [90/2297856] via 172.20.12.1, 00:41:29, Serial1/0
      100.0.0.0/30 is subnetted, 1 subnets
```



```
C        172.20.2.0/24 is directly connected, FastEthernet0/0
L        172.20.2.1/32 is directly connected, FastEthernet0/0
D        172.20.11.0/28 [90/2681856] via 172.20.12.1, 00:43:01, Serial1/0
C        172.20.12.0/30 is directly connected, Serial1/0
L        172.20.12.2/32 is directly connected, Serial1/0
```

Explanation

HQ Router – EIGRP AS 200 Routes

- Routes such as `10.20.41.0/24`, `10.20.89.0/24`, and `10.20.97.0/24` are learned via **BGP**, but they **originate from EIGRP AS200**, redistributed into BGP and sent to HQ.
- This confirms successful **route redistribution from EIGRP AS200 into BGP**, and the HQ router is receiving them via `10.100.1.1.`

ER2 Router – EIGRP AS 100 Routes

- Routes marked **EX (external EIGRP)** like `10.10.0.0/24`, `10.20.2.0/24`, etc., are learned via **EIGRP AS100** and redistributed into ER2 through ASBR.
- Routes are received via `172.20.12.1`, confirming that **EIGRP AS100 is correctly redistributing** into another domain (likely via BGP or mutual redistribution).

Output/Result [EIGRP – OSPF – BGP]: [6m]

OR2 - [6m] – highlight network routes from EIGRP AS200. EIGRP AS100 and BGP

EIGRP AS 200:

```
O E2     10.10.2.0/24 [110/20] via 172.21.12.1, 00:48:52, FastEthernet0/0
O E1     10.20.41.0/24 [110/85] via 172.21.12.1, 00:48:52, FastEthernet0/0
O E1     10.20.89.0/24 [110/85] via 172.21.12.1, 00:48:52, FastEthernet0/0
O E1     10.20.97.0/24 [110/85] via 172.21.12.1, 00:48:52, FastEthernet0/0
O E1     10.30.16.0/20 [110/85] via 172.21.12.1, 00:14:55, FastEthernet0/0
O IA     10.30.16.1/32 [110/65] via 172.23.23.3, 00:16:14, Serial1/0
```

```
O E2     100.100.102.0 [110/20] via 172.21.12.1, 00:48:52, FastEthernet0/0
      172.20.0.0/16 is variably subnetted, 3 subnets, 3 masks
O E1     172.20.2.0/24 [110/85] via 172.21.12.1, 00:48:52, FastEthernet0/0
O E2     172.20.11.0/28 [110/20] via 172.21.12.1, 00:48:52, FastEthernet0/0
```

EIGRP AS 100:

```
O E2     10.10.1.0/24 [110/20] via 172.21.12.1, 00:48:52, FastEthernet0/0
O E2     10.10.2.0/24 [110/20] via 172.21.12.1, 00:48:52, FastEthernet0/0
O E1     10.20.41.0/24 [110/85] via 172.21.12.1, 00:48:52, FastEthernet0/0
O E1     10.20.89.0/24 [110/85] via 172.21.12.1, 00:48:52, FastEthernet0/0
```

```
O IA     10.30.18.1/32 [110/65] via 172.23.23.3, 00:26:26, Serial1/0
O E2     10.100.1.0/24 [110/20] via 172.21.12.1, 00:59:04, FastEthernet0/0
      100.0.0.0/30 is subnetted, 1 subnets
O E2     100.100.102.0 [110/20] via 172.21.12.1, 00:59:04, FastEthernet0/0
      172.20.0.0/16 is variably subnetted, 3 subnets, 3 masks
```

```
O E2     172.20.11.0/28 [110/20] via 172.21.12.1, 00:59:04, FastEthernet0/0
O E2     172.20.12.0/30 [110/20] via 172.21.12.1, 00:59:04, FastEthernet0/0
      172.21.0.0/16 is variably subnetted, 3 subnets, 3 masks
O        172.21.11.0/28 [110/65] via 172.21.12.1, 00:59:04, FastEthernet0/0
```

BGP:

```
O E2     192.121.11.0 [110/1] via 172.21.12.1, 00:58:35, FastEthernet0/0
      192.122.12.0/30 is subnetted, 1 subnets
O E2     192.122.12.0 [110/1] via 172.21.12.1, 00:58:35, FastEthernet0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, Loopback2
L        192.168.2.1/32 is directly connected, Loopback2
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected, Loopback3
L        192.168.3.1/32 is directly connected, Loopback3
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/24 is directly connected, Loopback4
L        192.168.4.1/32 is directly connected, Loopback4
O E2  192.168.10.0/24 [110/1] via 172.21.12.1, 00:58:35, FastEthernet0/0
O E2  192.168.11.0/24 [110/1] via 172.21.12.1, 00:58:35, FastEthernet0/0
O E2  192.168.20.0/24 [110/1] via 172.21.12.1, 00:58:35, FastEthernet0/0
O E2  192.168.30.0/24 [110/1] via 172.21.12.1, 00:58:35, FastEthernet0/0
O E2  200.100.20.0/24 [110/1] via 172.21.12.1, 00:58:19, FastEthernet0/0
      201.110.11.0/28 is subnetted, 1 subnets
O E2     201.110.11.0 [110/1] via 172.21.12.1, 00:58:35, FastEthernet0/0
      201.120.12.0/30 is subnetted, 1 subnets
O E2     201.120.12.0 [110/1] via 172.21.12.1, 00:58:35, FastEthernet0/0
      201.122.12.0/30 is subnetted, 1 subnets
O        201.122.12.0 [110/65] via 172.21.12.1, 00:59:04, FastEthernet0/0
      201.223.23.0/29 is subnetted, 1 subnets
O E2     201.223.23.0 [110/1] via 172.21.12.1, 00:58:19, FastEthernet0/0
      201.223.32.0/29 is subnetted, 1 subnets
O E2     201.223.32.0 [110/1] via 172.21.12.1, 00:58:19, FastEthernet0/0
      202.230.23.0/29 is subnetted, 1 subnets
O E2     202.230.23.0 [110/1] via 172.21.12.1, 00:58:35, FastEthernet0/0
O E2  203.23.30.0/24 [110/1] via 172.21.12.1, 00:58:35, FastEthernet0/0
O E2  210.11.1.0/24 [110/1] via 172.21.12.1, 00:58:35, FastEthernet0/0
O E2  210.11.11.0/24 [110/1] via 172.21.12.1, 00:58:35, FastEthernet0/0
O E2  210.200.20.0/24 [110/1] via 172.21.12.1, 00:58:19, FastEthernet0/0
O E2  220.202.20.0/24 [110/1] via 172.21.12.1, 00:58:19, FastEthernet0/0
```

#8: Summarization - #`show ip route`

Screenshot the output then highlight the particular output and explain it.

OR3 – output before and after summarization for Loopback OR3 and Loopback ER1

Before:

```
O E2     10.10.0.0/16 [110/20] via 172.23.23.2, 00:06:42, Serial1/0
O E2     10.10.1.0/24 [110/20] via 172.23.23.2, 00:06:42, Serial1/0
O E2     10.10.2.0/24 [110/20] via 172.23.23.2, 00:06:42, Serial1/0
O E1     10.20.41.0/24 [110/149] via 172.23.23.2, 00:06:42, Serial1/0
O E1     10.20.89.0/24 [110/149] via 172.23.23.2, 00:06:42, Serial1/0
O E1     10.20.97.0/24 [110/149] via 172.23.23.2, 00:06:42, Serial1/0
C        10.30.16.0/24 is directly connected, Loopback16
L        10.30.16.1/32 is directly connected, Loopback16
C        10.30.17.0/24 is directly connected, Loopback17
L        10.30.17.1/32 is directly connected, Loopback17
C        10.30.18.0/24 is directly connected, Loopback18
L        10.30.18.1/32 is directly connected, Loopback18
O E2     10.100.1.0/24 [110/20] via 172.23.23.2, 00:06:42, Serial1/0
```

After:

```
O E2     10.10.0.0/16 [110/20] via 172.23.23.2, 00:32:55, Serial1/0
O E2     10.10.1.0/24 [110/20] via 172.23.23.2, 01:00:06, Serial1/0
O E2     10.10.2.0/24 [110/20] via 172.23.23.2, 01:00:06, Serial1/0
O E1     10.20.41.0/24 [110/149] via 172.23.23.2, 01:00:06, Serial1/0
O E1     10.20.89.0/24 [110/149] via 172.23.23.2, 01:00:06, Serial1/0
O E1     10.20.97.0/24 [110/149] via 172.23.23.2, 01:00:06, Serial1/0
O        10.30.0.0/16 is a summary, 01:00:06, Null0
O E2     10.30.16.0/20 [110/20] via 172.23.23.2, 00:32:55, Serial1/0
C        10.30.16.0/24 is directly connected, Loopback16
L        10.30.16.1/32 is directly connected, Loopback16
C        10.30.17.0/24 is directly connected, Loopback17
L        10.30.17.1/32 is directly connected, Loopback17
C        10.30.18.0/24 is directly connected, Loopback18
L        10.30.18.1/32 is directly connected, Loopback18
```

Output on HQ before and after summarize loopback ER1 and OR3:

ER1 - output before and after summarization for Loopback OR3 and Loopback ER1

Before:

```
D        10.10.0.0/16 is a summary, 00:03:41, Null0
C        10.10.1.0/24 is directly connected, Loopback1
L        10.10.1.1/32 is directly connected, Loopback1
C        10.10.2.0/24 is directly connected, Loopback2
L        10.10.2.1/32 is directly connected, Loopback2
D        10.20.41.0/24 [90/2297856] via 172.20.12.2, 00:03:41, Serial5/2
D        10.20.89.0/24 [90/2297856] via 172.20.12.2, 00:03:41, Serial5/2
D        10.20.97.0/24 [90/2297856] via 172.20.12.2, 00:03:41, Serial5/2
O IA     10.30.16.1/32 [110/130] via 172.21.11.1, 00:00:09, Serial5/0
O IA     10.30.17.1/32 [110/130] via 172.21.11.1, 00:00:09, Serial5/0
O IA     10.30.18.1/32 [110/130] via 172.21.11.1, 00:00:09, Serial5/0
C        10.100.1.0/24 is directly connected, Loopback0
```

After:

```
O IA     10.3.3.1/32 [110/130] via 172.21.11.1, 01:22:57, Serial5/0
D        10.10.0.0/16 is a summary, 01:22:57, Null0
C        10.10.1.0/24 is directly connected, Loopback1
L        10.10.1.1/32 is directly connected, Loopback1
C        10.10.2.0/24 is directly connected, Loopback2
L        10.10.2.1/32 is directly connected, Loopback2
D        10.20.41.0/24 [90/2297856] via 172.20.12.2, 01:22:57, Serial5/2
D        10.20.89.0/24 [90/2297856] via 172.20.12.2, 01:22:57, Serial5/2
D        10.20.97.0/24 [90/2297856] via 172.20.12.2, 01:22:57, Serial5/2
O IA     10.30.0.0/16 [110/130] via 172.21.11.1, 01:05:05, Serial5/0
D EX     10.30.16.0/20 [170/2195456] via 172.20.12.2, 01:02:40, Serial5/2
C        10.100.1.0/24 is directly connected, Loopback0
```

Before:

```
B        10.10.0.0/16 [20/128256] via 10.100.1.1, 00:01:46
B        10.10.1.0/24 [20/0] via 10.100.1.1, 00:01:46
B        10.10.2.0/24 [20/0] via 10.100.1.1, 00:01:46
B        10.20.41.0/24 [20/2297856] via 10.100.1.1, 00:01:46
B        10.20.89.0/24 [20/2297856] via 10.100.1.1, 00:01:46
B        10.20.97.0/24 [20/2297856] via 10.100.1.1, 00:01:46
B        10.30.16.0/20 [20/0] via 210.11.11.1, 00:00:18
B        10.30.16.1/32 [20/130] via 10.100.1.1, 00:00:35
B        10.30.17.1/32 [20/130] via 10.100.1.1, 00:00:35
B        10.30.18.1/32 [20/130] via 10.100.1.1, 00:00:35
B        10.100.1.0/24 [20/0] via 10.100.1.1, 00:01:46
```

After:

```
B        10.10.0.0/16 [20/128256] via 10.100.1.1, 01:03:06
B        10.10.1.0/24 [20/0] via 10.100.1.1, 01:03:06
B        10.10.2.0/24 [20/0] via 10.100.1.1, 01:03:06
B        10.20.41.0/24 [20/2297856] via 10.100.1.1, 01:03:06
B        10.20.89.0/24 [20/2297856] via 10.100.1.1, 01:03:06
B        10.20.97.0/24 [20/2297856] via 10.100.1.1, 01:03:06
B        10.30.0.0/16 [20/130] via 10.100.1.1, 00:49:16
B        10.30.16.0/20 [20/2195456] via 10.100.1.1, 00:47:44
B        10.100.1.0/24 [20/0] via 10.100.1.1, 01:03:06
```

OR1

Before

```
O E2     10.10.0.0/16 [110/20] via 172.21.11.11, 00:08:08, Serial1/0
O E2     10.10.1.0/24 [110/20] via 172.21.11.11, 00:08:08, Serial1/0
O E2     10.10.2.0/24 [110/20] via 172.21.11.11, 00:08:08, Serial1/0
O E1     10.20.41.0/24 [110/84] via 172.21.11.11, 00:08:08, Serial1/0
O E1     10.20.89.0/24 [110/84] via 172.21.11.11, 00:08:08, Serial1/0
O E1     10.20.97.0/24 [110/84] via 172.21.11.11, 00:08:08, Serial1/0
O IA     10.30.16.1/32 [110/66] via 172.21.12.2, 00:02:28, FastEthernet0/0
O IA     10.30.17.1/32 [110/66] via 172.21.12.2, 00:02:28, FastEthernet0/0
O IA     10.30.18.1/32 [110/66] via 172.21.12.2, 00:02:28, FastEthernet0/0
O E2     10.100.1.0/24 [110/20] via 172.21.11.11, 00:08:08, Serial1/0
```

After

```
O E2     10.10.0.0/16 [110/20] via 172.21.11.11, 00:25:16, Serial1/0
O E2     10.10.1.0/24 [110/20] via 172.21.11.11, 01:04:02, Serial1/0
O E2     10.10.2.0/24 [110/20] via 172.21.11.11, 01:04:02, Serial1/0
O E1     10.20.41.0/24 [110/84] via 172.21.11.11, 01:04:02, Serial1/0
O E1     10.20.89.0/24 [110/84] via 172.21.11.11, 01:04:02, Serial1/0
O E1     10.20.97.0/24 [110/84] via 172.21.11.11, 01:04:02, Serial1/0
O IA     10.30.0.0/16 [110/66] via 172.21.12.2, 00:45:57, FastEthernet0/0
O E2     10.30.16.0/20 [110/20] via 172.21.11.11, 00:25:16, Serial1/0
O E2     10.100.1.0/24 [110/20] via 172.21.11.11, 01:04:02, Serial1/0
```

ER2 -

Before

```
D        10.10.0.0/16 [90/2297856] via 172.20.12.1, 00:08:48, Serial1/0
D EX     10.10.1.0/24 [170/2195456] via 172.20.12.1, 00:08:48, Serial1/0
D EX     10.10.2.0/24 [170/2195456] via 172.20.12.1, 00:08:48, Serial1/0
C        10.20.41.0/24 is directly connected, Loopback2
L        10.20.41.1/32 is directly connected, Loopback2
C        10.20.89.0/24 is directly connected, Loopback3
L        10.20.89.1/32 is directly connected, Loopback3
C        10.20.97.0/24 is directly connected, Loopback1
L        10.20.97.1/32 is directly connected, Loopback1
D EX     10.30.16.1/32 [170/2195456] via 172.20.12.1, 00:02:58, Serial1/0
D EX     10.30.17.1/32 [170/2195456] via 172.20.12.1, 00:02:58, Serial1/0
D EX     10.30.18.1/32 [170/2195456] via 172.20.12.1, 00:02:58, Serial1/0
D        10.100.1.0/24 [90/2297856] via 172.20.12.1, 00:08:48, Serial1/0
```

After

```
D        10.10.0.0/16 [90/2297856] via 172.20.12.1, 01:16:06, Serial1/0
D EX     10.10.1.0/24 [170/2195456] via 172.20.12.1, 01:16:06, Serial1/0
D EX     10.10.2.0/24 [170/2195456] via 172.20.12.1, 01:16:06, Serial1/0
C        10.20.41.0/24 is directly connected, Loopback2
L        10.20.41.1/32 is directly connected, Loopback2
C        10.20.89.0/24 is directly connected, Loopback3
L        10.20.89.1/32 is directly connected, Loopback3
C        10.20.97.0/24 is directly connected, Loopback1
L        10.20.97.1/32 is directly connected, Loopback1
D EX     10.30.0.0/16 [170/2195456] via 172.20.12.1, 00:58:04, Serial1/0
D EX     10.30.16.0/20 [170/2195456] via 172.20.12.1, 00:55:39, Serial1/0
D        10.100.1.0/24 [90/2297856] via 172.20.12.1, 01:16:06, Serial1/0
```

Test Connection:

Ping from PC1 to 10.30.18.1 (Loopback on OR3)

```
VPCS> ping 10.30.18.1

84 bytes from 10.30.18.1 icmp_seq=1 ttl=251 time=156.498 ms
84 bytes from 10.30.18.1 icmp_seq=2 ttl=251 time=243.823 ms
84 bytes from 10.30.18.1 icmp_seq=3 ttl=251 time=69.406 ms
84 bytes from 10.30.18.1 icmp_seq=4 ttl=251 time=114.320 ms
84 bytes from 10.30.18.1 icmp_seq=5 ttl=251 time=132.338 ms
```

Ping from PC2 to 10.10.1.1 (Loopback on ER1)

```
VPCS> ping 10.10.1.1

84 bytes from 10.10.1.1 icmp_seq=1 ttl=254 time=19.953 ms
84 bytes from 10.10.1.1 icmp_seq=2 ttl=254 time=15.830 ms
84 bytes from 10.10.1.1 icmp_seq=3 ttl=254 time=28.030 ms
84 bytes from 10.10.1.1 icmp_seq=4 ttl=254 time=22.985 ms
84 bytes from 10.10.1.1 icmp_seq=5 ttl=254 time=117.814 ms
```

Explanation:

After the summarization process, we can see significant improvements in the routing tables across the network devices. Initially, each router displayed multiple individual routes for the loopback interfaces, such as `/24` and `/32` prefixes. This resulted in larger routing tables and increased the amount of routing information exchanged between routers. After applying summarization, these specific routes were aggregated into summarized networks, such as `10.10.0.0/16` and `10.30.16.0/20`. This effectively reduced the number of entries in the routing tables, making them more concise and manageable. Summarization not only simplifies the routing process but also minimizes the size of routing updates, leading to reduced bandwidth usage and improved router efficiency. The connectivity was verified through successful ping tests to the loopback addresses, confirming that summarization maintained proper reachability while optimizing the network.

#9: GRE Tunnel

ER1 – `show ip interface brief` and test connection (ping to tunnel ip (100.100.100.2) on ISP2); `show ip route`

ISP2 – `show ip route` (highlight static route)

sh ip int br:

```
ER1#sh ip int br
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            unassigned      YES NVRAM  administratively down down
FastEthernet1/0            unassigned      YES NVRAM  administratively down down
FastEthernet1/1            unassigned      YES NVRAM  administratively down down
FastEthernet2/0            unassigned      YES NVRAM  administratively down down
FastEthernet2/1            unassigned      YES NVRAM  administratively down down
GigabitEthernet3/0         unassigned      YES NVRAM  administratively down down
Serial5/0                  172.21.11.11    YES NVRAM  up                    up
Serial5/1                  172.20.11.11    YES NVRAM  up                    up
Serial5/2                  172.20.12.1     YES NVRAM  up                    up
Serial5/3                  unassigned      YES NVRAM  administratively down down
Serial6/0                  unassigned      YES NVRAM  administratively down down
Serial6/1                  unassigned      YES NVRAM  administratively down down
Serial6/2                  unassigned      YES NVRAM  administratively down down
Serial6/3                  unassigned      YES NVRAM  administratively down down
Loopback0                  10.100.1.1      YES NVRAM  up                    up
Loopback1                  10.10.1.1       YES NVRAM  up                    up
Loopback2                  10.10.2.1       YES NVRAM  up                    up
Tunnel0                    100.100.102.1   YES NVRAM  up                    up
```

ping to tunnel ip on ISP2:

```
ER1#ping 100.100.102.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.100.102.2, timeout is 2 seconds:
!!!!!
```

Show ip route:

```
O IA     10.30.16.0/20 [110/130] via 172.21.11.1, 00:04:17, Serial5/0
C        10.100.1.0/24 is directly connected, Loopback0
L        10.100.1.1/32 is directly connected, Loopback0
      100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        100.100.102.0/30 is directly connected, Tunnel0
L        100.100.102.1/32 is directly connected, Tunnel0
      172.20.0.0/16 is variably subnetted, 5 subnets, 4 masks
D        172.20.2.0/24 [90/2172416] via 172.20.12.2, 00:05:01, Serial5/2
C        172.20.11.0/28 is directly connected, Serial5/1
```

```
B        10.10.1.0/24 [20/0] via 210.11.1.1, 00:03:31
S        10.10.1.1/32 [1/0] via 100.100.102.1
B        10.10.2.0/24 [20/0] via 210.11.1.1, 00:03:31
S        10.20.1.1/32 [1/0] via 100.100.102.1
B        10.20.41.0/24 [20/0] via 210.11.1.1, 00:04:02
B        10.20.89.0/24 [20/0] via 210.11.1.1, 00:04:02
B        10.20.97.0/24 [20/0] via 210.11.1.1, 00:04:02
B        10.30.16.0/20 [20/0] via 210.11.1.1, 00:04:02
S        10.100.1.1/32 is directly connected
                       is directly connected, Serial3/1
```

Test connection – ping 220.202.20.1 source 10.10.1.1

```
ER1# ping 220.202.20.1 source 10.10.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 220.202.20.1, timeout is 2 seconds:
Packet sent with a source address of 10.10.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/36/64 ms
```

Explanation:

The tunnel interface Tunnel0 on ER1 shows IP `100.100.102.1` is up/up. and pings to ISP 2 are successful. confirming that the GRE tunnel is established correctly.

On ISP 2, the routing table includes static routes pointing to networks like `10.10.1.0/24` and `/32` via `100.100.102.1` and interface se3/1 is directly connected to `10.100.1.1/32`, proving that the GRE tunnel traffic is being routed through the correct path.

#10: Path Manipulation and Route Filtering

Write the policy/configuration command (You may copy from running file @ verify the configured policy.

On BG3:

BG3(config)#route-map PREFER-ISP2 permit 10

BG3(config-route-map)#set local-preference 200

BG3(config)#router bgp 300

BG3(config-router)#neighbor 220.202.20.1 route-map PREFER-ISP2 in

Examine output **before apply the policy**: (You may snapshot from the simulation-GNS3 or PT).) Use the verification command.

Ping and traceroute from **PC2 – User 1** (203.23.30.30)	Ping and traceroute from **User 1 – PC2** (172.20.2.xxx) **[User 1;BG3 to PC2;ER2]**

```
VPCS> ping 203.23.30.30

84 bytes from 203.23.30.30 icmp_seq=1 ttl=57 time=171.445 ms
84 bytes from 203.23.30.30 icmp_seq=2 ttl=57 time=218.969 ms
84 bytes from 203.23.30.30 icmp_seq=3 ttl=47 time=168.570 ms
84 bytes from 203.23.30.30 icmp_seq=4 ttl=57 time=167.911 ms
84 bytes from 203.23.30.30 icmp_seq=5 ttl=41 time=225.396 ms
```

```
VPCS> trace 203.23.30.30
trace to 203.23.30.30, 8 hops max, press Ctrl+C to stop
 1   172.20.2.1   12.122 ms  11.892 ms  9.460 ms
 2   172.20.12.1   42.816 ms  30.623 ms  31.453 ms
 3   172.20.11.1   50.027 ms  31.493 ms  29.587 ms
 4   201.110.11.11   74.452 ms  43.161 ms  54.105 ms
 5   192.121.11.1   79.714 ms  86.680 ms  *
 6   192.122.12.2   137.974 ms  78.391 ms  68.871 ms
 7   192.122.12.1   137.486 ms  81.120 ms  88.058 ms
 8     **203.23.30.30   96.679 ms (ICMP type:3, code:3, Destination port
 ms
```

```
VPCS> ping 172.20.2.11

84 bytes from 172.20.2.11 icmp_seq=1 ttl=43 time=200.810 ms
84 bytes from 172.20.2.11 icmp_seq=2 ttl=57 time=200.897 ms
84 bytes from 172.20.2.11 icmp_seq=3 ttl=49 time=178.776 ms
84 bytes from 172.20.2.11 icmp_seq=4 ttl=37 time=367.903 ms
84 bytes from 172.20.2.11 icmp_seq=5 ttl=51 time=142.203 ms
```

```
VPCS> trace 172.20.2.11
trace to 172.20.2.11, 8 hops max, press Ctrl+C to stop
 1   203.23.30.1   8.430 ms  10.045 ms  11.032 ms
 2   202.230.23.2   32.073 ms  31.103 ms  33.887 ms
 3   192.122.12.1   56.210 ms  121.245 ms  43.003 ms
 4   192.121.11.11   51.083 ms  70.053 ms  160.417 ms
 5   201.110.11.1   90.413 ms  78.665 ms  206.861 ms
 6   172.20.11.11   118.989 ms  180.369 ms  103.076 ms
 7   172.20.12.2   282.996 ms  226.551 ms
```

S

BG3 – show ip bgp (highlight route 172.20.2.0)

```
BG3#sh ip bgp 172.20.2.0
BGP routing table entry for 172.20.2.0/24, version 6525
Paths: (2 available, best #2, table default)
  Advertised to update-groups:
     1
  Refresh Epoch 3
  200 65100 65010
    220.202.20.1 from 220.202.20.1 (220.202.20.1)
      Origin incomplete, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
  Refresh Epoch 1
  100
    192.168.20.1 from 192.168.20.1 (192.168.20.1)
      Origin incomplete, metric 286720, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
```

Output/Result/ Test Connection **(After apply the policy)**: (You may snapshot from the simulation-GNS3 or PT). Use the verification command and explain it.

Ping and traceroute from **PC2 – User 1** (203.23.30.30)

```
VPCS> ping 203.23.30.30

84 bytes from 203.23.30.30 icmp_seq=1 ttl=57 time=220.548 ms
84 bytes from 203.23.30.30 icmp_seq=2 ttl=55 time=208.171 ms
84 bytes from 203.23.30.30 icmp_seq=3 ttl=57 time=137.359 ms
84 bytes from 203.23.30.30 icmp_seq=4 ttl=57 time=151.757 ms
```

```
VPCS> tracer 203.23.30.30
trace to 203.23.30.30, 8 hops max, press Ctrl+C to stop
 1   172.20.2.1   12.296 ms  9.183 ms  9.642 ms
 2   172.20.12.1   33.311 ms  29.711 ms  34.255 ms
 3   172.20.11.1   93.907 ms  46.142 ms  33.105 ms
 4   201.110.11.11   274.688 ms  72.331 ms  73.285 ms
 5   192.121.11.1   73.372 ms  111.958 ms  *
 6   192.122.12.2   77.200 ms  104.998 ms  84.648 ms
 7   202.230.23.3   102.214 ms  125.631 ms  116.611 ms
 8   192.122.12.2   113.610 ms  357.684 ms
```

Ping and traceroute from **User 1 – PC2** (172.20.2.xxx)

```
VPCS> ping 172.20.2.11

84 bytes from 172.20.2.11 icmp_seq=1 ttl=57 time=107.580 ms
84 bytes from 172.20.2.11 icmp_seq=2 ttl=57 time=106.559 ms
84 bytes from 172.20.2.11 icmp_seq=3 ttl=57 time=135.388 ms
84 bytes from 172.20.2.11 icmp_seq=4 ttl=55 time=137.927 ms
84 bytes from 172.20.2.11 icmp_seq=5 ttl=57 time=179.559 ms
```

```
VPCS> tracer 172.20.2.11
trace to 172.20.2.11, 8 hops max, press Ctrl+C to stop
 1   203.23.30.1   5.856 ms  2.190 ms  6.947 ms
 2   201.223.32.2   45.662 ms  30.970 ms  33.192 ms
 3   201.120.12.1   73.119 ms  61.322 ms  61.766 ms
 4   172.20.11.11   77.246 ms  253.021 ms  90.578 ms
 5   172.20.12.2   113.357 ms  95.959 ms  173.653 ms
 6   *172.20.2.11   101.659 ms (ICMP type:3, code:3, Destination port unreachable)
```

BG3 – show ip bgp (highlight route 172.20.2.0)

```
BG3#sh ip bgp 172.20.2.0
BGP routing table entry for 172.20.2.0/24, version 6854
Paths: (2 available, best #1, table default)
  Advertised to update-groups:
     1
  Refresh Epoch 4
  200 65100 65010
    220.202.20.1 from 220.202.20.1 (220.202.20.1)
      Origin incomplete, localpref 200, valid, external, best
      rx pathid: 0, tx pathid: 0x0
  Refresh Epoch 1
  100 65100 65010
    192.168.20.1 from 192.168.20.1 (192.168.20.1)
      Origin incomplete, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
```

Explanation:

Before Applying Policy (Route Preference Output)

- The `show ip bgp 172.20.2.0` output shows **192.168.20.1** path is selected as **best** (default local-preference 100).
- Traceroute from PC2 (203.23.30.30) and User 1 (172.20.2.x) shows traffic going via **ISP1 path**.

After Applying Route-Map Policy on BG3

A **route-map with local-preference 200** is applied to **ISP2 (220.202.20.1)** neighbor, making it more preferred.

Updated `show ip bgp` confirms **220.202.20.1 path** is now selected as **best**, overriding the previous path due to higher local preference.

Traceroute results from both PC2 and User 1 now show traffic is going via **ISP2 path**, verifying that **path manipulation is successful**.

#11: PBR and Route Filtering Configuration - #`show ip bgp`**/**`ping and traceroute`

Examine output **BEFORE apply the policy for PBR**: (You may snapshot from the simulation-GNS3 or PT) Use the verification command.

```
BG3#sh ip bgp 10.2.2.1
BGP routing table entry for 10.2.2.1/32, version 29
Paths: (2 available, best #2, table default)
  Advertised to update-groups:
     1
  Refresh Epoch 1
  100 65100 65010
    192.168.20.1 from 192.168.20.1 (192.168.20.1)
      Origin incomplete, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
  Refresh Epoch 1
  200 65100 65010
    220.202.20.1 from 220.202.20.1 (220.202.20.1)
      Origin incomplete, localpref 200, valid, external, best
      rx pathid: 0, tx pathid: 0x0
```

Verify current path from Loopback 2/3/4 OR2 – User 1; ping 203.23.30.30 source Lo2.

```
OR2#ping 203.23.30.30 source loopback2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.23.30.30, timeout is 2 seconds:
Packet sent with a source address of 192.168.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 112/140/180 ms
OR2#
```

Verify current path from Loopback 2/3/4 OR2 – User 1; traceroute 203.23.30.30 source Lo2. State the current path

```
OR2#traceroute 203.23.30.30 source loopback2
Type escape sequence to abort.
Tracing the route to 203.23.30.30
VRF info: (vrf in name/id, vrf out name/id)
  1 172.21.12.1 12 msec 28 msec 20 msec
  2 172.21.11.11 48 msec 12 msec 24 msec
  3 172.20.11.1 60 msec 52 msec 52 msec
  4 201.110.11.11 80 msec 48 msec 64 msec
  5 192.121.11.1 40 msec *  100 msec
  6 192.122.12.2 112 msec 292 msec 120 msec
  7 202.230.23.3 104 msec
    192.122.12.1 220 msec
    202.230.23.3 132 msec
  8  *
    203.23.30.30 156 msec *
```

Examine output **AFTER apply the policy for PBR**: (You may snapshot from the simulation-GNS3 or PT). Use the verification command.

Verify current path from Loopback 2/3/4 OR2 – User 1; ping 203.23.30.30 source Lo2.

```
OR2#ping 203.23.30.30 source loopback2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.23.30.30, timeout is 2 seconds:
Packet sent with a source address of 192.168.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 112/174/240 ms
```

Verify current path from Loopback 2/3/4 OR2 – User 1; traceroute 203.23.30.30 source Lo2. State the current path

```
OR2#traceroute 203.23.30.30 source loopback2
Type escape sequence to abort.
Tracing the route to 203.23.30.30
VRF info: (vrf in name/id, vrf out name/id)
  1 172.21.12.1 8 msec 20 msec 20 msec
  2 172.21.11.11 48 msec 16 msec 36 msec
  3 100.100.102.2 212 msec 80 msec 92 msec
  4 201.223.32.3 80 msec 288 msec 80 msec
  5 203.23.30.30 248 msec 88 msec 188 msec
```

Explanation: – `show ip access-list / show ip prefix-list` & **show route-map**

```
ER1#sh ip access-lists
Extended IP access list 101
    10 permit ip host 192.168.2.1 203.23.30.0 0.0.0.255 (17 matches)
    20 permit ip host 192.168.3.1 203.23.30.0 0.0.0.255
    30 permit ip host 192.168.4.1 203.23.30.0 0.0.0.255
```

```
route-map PBR-TUNNEL, permit, sequence 10
  Match clauses:
    ip address (access-lists): 101
  Set clauses:
    ip next-hop 100.100.102.2
    interface Tunnel0
  Policy routing matches: 17 packets, 904 bytes
```

#12: DHCP Configuration - #`show ip dhcp pool / show ip dhcp binding`

DHCP Pool – `show ip dhcp pool / show ip dhcp binding`

ER2

```
ER2#sh ip dhcp pool

Pool DHCP-USER-POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 172.20.2.1           172.20.2.1       - 172.20.2.254      0
```

```
ER2#sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
172.20.2.11         0100.5079.6668.00       Jul 14 2025 01:54 AM    Automatic
172.20.2.12         0100.5079.6668.02       Jul 14 2025 01:54 AM    Automatic
```

IP Assign to PC1 & PC2

PC1:

```
VPCS> dhcp
DDORA IP 172.20.2.11/24 GW 172.20.2.1
```

PC2:

```
VPCS> dhcp
DDORA IP 172.20.2.12/24 GW 172.20.2.1
```

Output/Result/ Test Connection: (

Ping PC1 – PC2

```
VPCS> ping 172.20.2.12 -c 5

84 bytes from 172.20.2.12 icmp_seq=1 ttl=64 time=0.131 ms
84 bytes from 172.20.2.12 icmp_seq=2 ttl=64 time=0.483 ms
84 bytes from 172.20.2.12 icmp_seq=3 ttl=64 time=0.706 ms
84 bytes from 172.20.2.12 icmp_seq=4 ttl=64 time=0.240 ms
84 bytes from 172.20.2.12 icmp_seq=5 ttl=64 time=0.725 ms
```

Ping PC2 – 210.11.1.1

```
VPCS> ping 210.11.1.1

84 bytes from 210.11.1.1 icmp_seq=1 ttl=253 time=55.101 ms
84 bytes from 210.11.1.1 icmp_seq=2 ttl=253 time=49.623 ms
84 bytes from 210.11.1.1 icmp_seq=3 ttl=253 time=54.108 ms
84 bytes from 210.11.1.1 icmp_seq=4 ttl=253 time=57.869 ms
84 bytes from 210.11.1.1 icmp_seq=5 ttl=253 time=77.084 ms
```

#13: Default Route Configuration

OR1 – `show ip route` and highlight the default route	**OR2 –** `show ip route` and highlight the default route

```
S*    0.0.0.0/0 [1/0] via 201.122.12.2
      10.0.0.0/8 is variably subnetted, 13 subnets, 4 masks
C        10.1.1.0/24 is directly connected, Loopback0
L        10.1.1.1/32 is directly connected, Loopback0
O        10.2.2.1/32 [110/2] via 172.21.12.2, 00:13:19, FastEthernet0/0
O        10.2.20.0/24 [110/2] via 172.21.12.2, 00:13:19, FastEthernet0/0
O IA     10.3.3.1/32 [110/66] via 172.21.12.2, 00:13:19, FastEthernet0/0
O E2     10.10.0.0/16 [110/20] via 172.21.11.11, 00:13:54, Serial1/0
O E2     10.10.1.0/24 [110/20] via 172.21.11.11, 00:13:54, Serial1/0
```

```
O*E1  0.0.0.0/0 [110/2] via 172.21.12.1, 00:18:31, FastEthernet0/0
      10.0.0.0/8 is variably subnetted, 14 subnets, 4 masks
O        10.1.1.1/32 [110/2] via 172.21.12.1, 00:18:31, FastEthernet0/0
C        10.2.2.0/24 is directly connected, Loopback0
L        10.2.2.1/32 is directly connected, Loopback0
C        10.2.20.0/24 is directly connected, FastEthernet2/0
L        10.2.20.1/32 is directly connected, FastEthernet2/0
O        10.3.3.1/32 [110/65] via 172.23.23.3, 00:19:11, Serial1/0
```

ER1 – `show ip route` and highlight the default route

```
O*E1  0.0.0.0/0 [110/65] via 172.21.11.1, 00:23:40, Serial5/0
      10.0.0.0/8 is variably subnetted, 15 subnets, 4 masks
O        10.1.1.1/32 [110/65] via 172.21.11.1, 00:23:40, Serial5/0
O        10.2.2.1/32 [110/66] via 172.21.11.1, 00:23:40, Serial5/0
O        10.2.20.0/24 [110/66] via 172.21.11.1, 00:23:40, Serial5/0
O IA     10.3.3.1/32 [110/130] via 172.21.11.1, 00:23:40, Serial5/0
D        10.10.0.0/16 is a summary, 00:23:40, Null0
C        10.10.1.0/24 is directly connected, Loopback1
L        10.10.1.1/32 is directly connected, Loopback1
```

Test the connection from User 2 (OR2) – `ping and trace`

Ping 201.122.12.1 & trace 201.122.12.1

```
User2> ping 201.122.12.1

84 bytes from 201.122.12.1 icmp_seq=1 ttl=254 time=28.829 ms
84 bytes from 201.122.12.1 icmp_seq=2 ttl=254 time=36.421 ms
84 bytes from 201.122.12.1 icmp_seq=3 ttl=254 time=23.272 ms
84 bytes from 201.122.12.1 icmp_seq=4 ttl=254 time=28.524 ms
84 bytes from 201.122.12.1 icmp_seq=5 ttl=254 time=19.093 ms
```

```
User2> trace 201.122.12.1
trace to 201.122.12.1, 8 hops max, press Ctrl+C to stop
 1   10.2.20.1   4.056 ms  9.572 ms  11.344 ms
 2   *172.21.12.1   30.549 ms (ICMP type:3, code:3, Destination port unreachable)  *
```

OR3 – **`show ip route`** and highlight the default route

```
O*E1  0.0.0.0/0 [110/66] via 172.23.23.2, 00:31:26, Serial1/0
      10.0.0.0/8 is variably subnetted, 19 subnets, 4 masks
O        10.1.1.1/32 [110/66] via 172.23.23.2, 00:31:26, Serial1/0
O        10.2.2.1/32 [110/65] via 172.23.23.2, 00:32:02, Serial1/0
O        10.2.20.0/24 [110/65] via 172.23.23.2, 00:32:02, Serial1/0
C        10.3.3.0/24 is directly connected, Loopback0
```

Explain the differences in the default route as seen in OR1, OR2, and OR3, and justify why these differences occur. **(5 m)**

The reason these differences occur is mainly because of how the default route is introduced into the network. OR1 has a static route manually set by the network admin, which means it always sends unknown traffic to a fixed next-hop IP without relying on any routing protocol.

Next, OR2 and OR3 learn their default routes through OSPF, but the type of OSPF route they get is different. OR2 receives an OSPF External Type 1 (E1) route, which takes into account both the internal OSPF cost and the external cost of the route. This is useful for making smarter routing decisions when there are multiple paths.

OR3, however, gets an External Type 2 (E2) route, where only the external cost is considered, and the internal OSPF path cost is ignored.

This setup could be intentional; possibly the admin wanted different behavior in different parts of the network, or it could be due to how ER1 is redistributing the default route on different interfaces or areas.

Explain why external type 1 is preferred for accessing to the external network

Type 1 external routes are preferred because they consider both the internal OSPF cost to reach the ASBR and the external cost to reach the destination. This gives a more accurate total path cost. In contrast, Type 2 routes only use the external cost, ignoring internal OSPF metrics, which can lead to suboptimal routing decisions when multiple ASBRs exist.

#14: EIGRP routes from AS 200 into OSPF as O E1

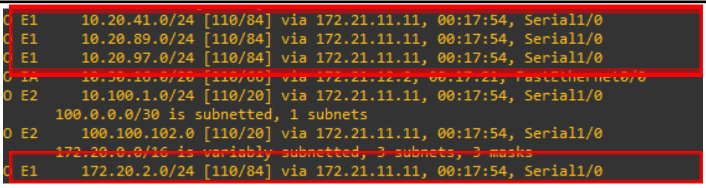
Provide the configuration and state which device should be configured

ER1Configuration:

ER1(config)#router ospf 10

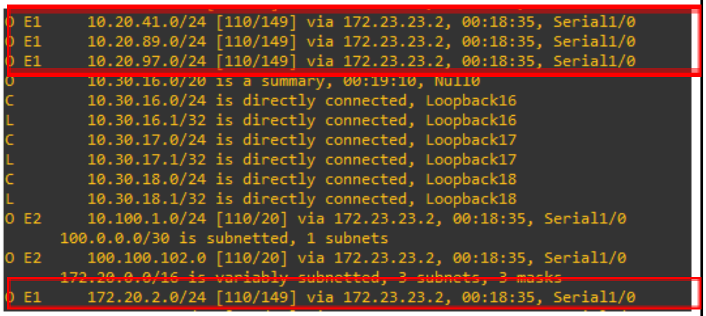
OR1 – `show ip route` and highlight the related routes

ER1(config)#no redistribute eigrp 200
ER1(config-router)#redistribute eigrp 200 metric-type 1 subnets



```
O E1     10.20.41.0/24 [110/84] via 172.21.11.11, 00:17:54, Serial1/0
O E1     10.20.89.0/24 [110/84] via 172.21.11.11, 00:17:54, Serial1/0
O E1     10.20.97.0/24 [110/84] via 172.21.11.11, 00:17:54, Serial1/0
O IA     10.30.16.0/20 [110/66] via 172.21.12.2, 00:17:21, FastEthernet0/0
O E2     10.100.1.0/24 [110/20] via 172.21.11.11, 00:17:54, Serial1/0
      100.0.0.0/30 is subnetted, 1 subnets
O E2     100.100.102.0 [110/20] via 172.21.11.11, 00:17:54, Serial1/0
      172.20.0.0/16 is variably subnetted, 3 subnets, 3 masks
O E1     172.20.2.0/24 [110/84] via 172.21.11.11, 00:17:54, Serial1/0
```

OR3 – `show ip route` and highlight the related routes



```
O E1     10.20.41.0/24 [110/149] via 172.23.23.2, 00:18:35, Serial1/0
O E1     10.20.89.0/24 [110/149] via 172.23.23.2, 00:18:35, Serial1/0
O E1     10.20.97.0/24 [110/149] via 172.23.23.2, 00:18:35, Serial1/0
O        10.30.16.0/20 is a summary, 00:19:10, Null0
C        10.30.16.0/24 is directly connected, Loopback16
L        10.30.16.1/32 is directly connected, Loopback16
C        10.30.17.0/24 is directly connected, Loopback17
L        10.30.17.1/32 is directly connected, Loopback17
C        10.30.18.0/24 is directly connected, Loopback18
L        10.30.18.1/32 is directly connected, Loopback18
O E2     10.100.1.0/24 [110/20] via 172.23.23.2, 00:18:35, Serial1/0
      100.0.0.0/30 is subnetted, 1 subnets
O E2     100.100.102.0 [110/20] via 172.23.23.2, 00:18:35, Serial1/0
      172.20.0.0/16 is variably subnetted, 3 subnets, 3 masks
O E1     172.20.2.0/24 [110/149] via 172.23.23.2, 00:18:35, Serial1/0
```

Explanation :
`show ip route` output on OR1 and OR3 shows routes from EIGRP AS 200 (e.g. `10.20.41.0/24`, `10.20.89.0/24`, `10.20.97.0/24`) marked as `O E1`. This indicates they were successfully redistributed into OSPF with external type 1 metric from ER1. The presence of `[110/84]` and `[110/149]` confirms these are external OSPF routes with correct cost values, and connectivity between domains is working as expected.

PART B

#1: Method for BGP routes update

1. Local Preference, used to inbound routing within AS
2. AS path prepending is used to influence inbound traffic by making a path less preferred.
3. Route Filtering, that use to controls routes accepted or not

#2: Purpose using loopback as neighbour ID – issues and solution

Issues:
BGP neighbor remains in Idle/Active state (sh ip bgp summary) because router cannot reach loopback of neighbor.

Solution:
Configure a static or IGP route to the neighbor's loopback, and set update-source + ebgp-multihop if needed.

#3: Local Preference attribute

BGP Local Preference Attribute, use to influence the path selection for outbound traffic within an Autonomous System.

Purpose: Determines the preferred exit point from the AS

Scope: It is propagated throughout the entire AS (iBGP only) and is not sent to external (eBGP) peers.

#4: Track path from User 2 to all loopback interfaces on ISP2

Track connectivity to all Loopback on ISP2

loopback 0: 200.100.20.1

```
User2> ping 200.100.20.1

84 bytes from 200.100.20.1 icmp_seq=1 ttl=251 time=312.870 ms
84 bytes from 200.100.20.1 icmp_seq=2 ttl=251 time=117.655 ms
84 bytes from 200.100.20.1 icmp_seq=3 ttl=251 time=110.162 ms
84 bytes from 200.100.20.1 icmp_seq=4 ttl=251 time=106.364 ms
84 bytes from 200.100.20.1 icmp_seq=5 ttl=251 time=137.204 ms

User2> trace 200.100.20.1
trace to 200.100.20.1, 8 hops max, press Ctrl+C to stop
 1   10.2.20.1   12.712 ms  10.967 ms  11.074 ms
 2   172.21.12.1   28.448 ms  27.845 ms  32.146 ms
 3   172.21.11.11   45.570 ms  46.932 ms  41.765 ms
 4   172.20.11.1   60.065 ms  49.058 ms  62.843 ms
 5   *201.120.12.2   94.918 ms (ICMP type:3, code:3, Destination port
```

loopback 1: 220.202.20.1

```
User2> ping 220.202.20.1

84 bytes from 220.202.20.1 icmp_seq=1 ttl=251 time=62.797 ms
84 bytes from 220.202.20.1 icmp_seq=2 ttl=251 time=45.691 ms
84 bytes from 220.202.20.1 icmp_seq=3 ttl=251 time=56.584 ms
84 bytes from 220.202.20.1 icmp_seq=4 ttl=251 time=88.011 ms
84 bytes from 220.202.20.1 icmp_seq=5 ttl=251 time=69.978 ms

User2> trace 220.202.20.1
trace to 220.202.20.1, 8 hops max, press Ctrl+C to stop
 1   10.2.20.1   2.041 ms  10.146 ms  10.398 ms
 2   172.21.12.1   35.796 ms  34.636 ms  31.138 ms
 3   172.21.11.11   50.296 ms  53.446 ms  54.006 ms
 4   172.20.11.1   65.138 ms  65.246 ms  71.010 ms
 5   *201.120.12.2   60.958 ms (ICMP type:3, code:3, Destination port
```

loopback 2: 210.200.20.1

```
User2> ping 210.200.20.1

84 bytes from 210.200.20.1 icmp_seq=1 ttl=251 time=123.037 ms
84 bytes from 210.200.20.1 icmp_seq=2 ttl=251 time=103.878 ms
84 bytes from 210.200.20.1 icmp_seq=3 ttl=251 time=116.116 ms
84 bytes from 210.200.20.1 icmp_seq=4 ttl=251 time=116.843 ms
84 bytes from 210.200.20.1 icmp_seq=5 ttl=251 time=111.480 ms

User2> trace 210.200.20.1
trace to 210.200.20.1, 8 hops max, press Ctrl+C to stop
 1   10.2.20.1   11.966 ms  10.430 ms  11.085 ms
 2   172.21.12.1   23.532 ms  20.424 ms  23.349 ms
 3   172.21.11.11   57.169 ms  51.054 ms  33.320 ms
 4   172.20.11.1   49.114 ms  52.974 ms  39.843 ms
 5   *201.120.12.2   56.454 ms (ICMP type:3, code:3, Destination port
```

Routing table on OR2

```
Gateway of last resort is 172.21.12.1 to network 0.0.0.0

O*E1  0.0.0.0/0 [110/2] via 172.21.12.1, 00:42:53, FastEthernet0/0
      10.0.0.0/8 is variably subnetted, 14 subnets, 4 masks
O        10.1.1.1/32 [110/2] via 172.21.12.1, 00:42:53, FastEthernet0/0
C        10.2.2.0/24 is directly connected, Loopback0
L        10.2.2.1/32 is directly connected, Loopback0
C        10.2.20.0/24 is directly connected, FastEthernet2/0
L        10.2.20.1/32 is directly connected, FastEthernet2/0
O        10.3.3.1/32 [110/65] via 172.23.23.3, 00:43:33, Serial1/0
O E2     10.10.0.0/16 [110/20] via 172.21.12.1, 00:42:53, FastEthernet0/0
O E2     10.10.1.0/24 [110/20] via 172.21.12.1, 00:42:53, FastEthernet0/0
O E2     10.10.2.0/24 [110/20] via 172.21.12.1, 00:42:53, FastEthernet0/0
O E1     10.20.41.0/24 [110/85] via 172.21.12.1, 00:42:53, FastEthernet0/0
O E1     10.20.89.0/24 [110/85] via 172.21.12.1, 00:42:53, FastEthernet0/0
O E1     10.20.97.0/24 [110/85] via 172.21.12.1, 00:42:53, FastEthernet0/0
O IA     10.30.16.0/20 [110/65] via 172.23.23.3, 00:43:15, Serial1/0
O E2     10.100.1.0/24 [110/20] via 172.21.12.1, 00:42:53, FastEthernet0/0
      100.0.0.0/30 is subnetted, 1 subnets
O E2     100.100.102.0 [110/20] via 172.21.12.1, 00:42:53, FastEthernet0/0
      172.20.0.0/16 is variably subnetted, 3 subnets, 3 masks
O E1     172.20.2.0/24 [110/85] via 172.21.12.1, 00:42:53, FastEthernet0/0
O E2     172.20.11.0/28 [110/20] via 172.21.12.1, 00:42:53, FastEthernet0/0
O E2     172.20.12.0/30 [110/20] via 172.21.12.1, 00:42:53, FastEthernet0/0
      172.21.0.0/16 is variably subnetted, 3 subnets, 3 masks
O        172.21.11.0/28 [110/65] via 172.21.12.1, 00:42:53, FastEthernet0/0
C        172.21.12.0/30 is directly connected, FastEthernet0/0
L        172.21.12.2/32 is directly connected, FastEthernet0/0
      172.23.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.23.23.0/29 is directly connected, Serial1/0
L        172.23.23.2/32 is directly connected, Serial1/0
      192.121.11.0/28 is subnetted, 1 subnets
O E2     192.121.11.0 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
      192.122.12.0/30 is subnetted, 1 subnets
O E2     192.122.12.0 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, Loopback2
L        192.168.2.1/32 is directly connected, Loopback2
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected, Loopback3
L        192.168.3.1/32 is directly connected, Loopback3
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/24 is directly connected, Loopback4
L        192.168.4.1/32 is directly connected, Loopback4
```

```
O E2  192.168.10.0/24 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
O E2  192.168.11.0/24 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
O E2  192.168.20.0/24 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
O E2  192.168.30.0/24 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
O E2  200.100.20.0/24 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
      201.110.11.0/28 is subnetted, 1 subnets
O E2     201.110.11.0 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
      201.120.12.0/30 is subnetted, 1 subnets
O E2     201.120.12.0 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
      201.122.12.0/30 is subnetted, 1 subnets
O        201.122.12.0 [110/65] via 172.21.12.1, 00:42:53, FastEthernet0/0
      201.223.23.0/29 is subnetted, 1 subnets
O E2     201.223.23.0 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
      201.223.32.0/29 is subnetted, 1 subnets
O E2     201.223.32.0 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
      202.230.23.0/29 is subnetted, 1 subnets
O E2     202.230.23.0 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
O E2  203.23.30.0/24 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
O E2  210.11.1.0/24 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
O E2  210.11.11.0/24 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
O E2  210.200.20.0/24 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
O E2  220.202.20.0/24 [110/1] via 172.21.12.1, 00:14:46, FastEthernet0/0
OR2#
```

Routing Table on OR1

Explanation based on the output

User 2 from subnet 172.20.2.0/24 can successfully ping all loopback interfaces on ISP2. This is because all ISP2 loopback IPs are already present in the routing tables of OR1 and OR2 via OSPF E2 routes, allowing

```
Gateway of last resort is 201.122.12.2 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 201.122.12.2
      10.0.0.0/8 is variably subnetted, 13 subnets, 4 masks
C        10.1.1.0/24 is directly connected, Loopback0
L        10.1.1.1/32 is directly connected, Loopback0
O        10.2.2.1/32 [110/2] via 172.21.12.2, 00:41:25, FastEthernet0/0
O        10.2.20.0/24 [110/2] via 172.21.12.2, 00:41:25, FastEthernet0/0
O IA     10.3.3.1/32 [110/66] via 172.21.12.2, 00:41:25, FastEthernet0/0
O E2     10.10.0.0/16 [110/20] via 172.21.11.11, 00:41:59, Serial1/0
O E2     10.10.1.0/24 [110/20] via 172.21.11.11, 00:41:59, Serial1/0
O E2     10.10.2.0/24 [110/20] via 172.21.11.11, 00:41:59, Serial1/0
O E1     10.20.41.0/24 [110/84] via 172.21.11.11, 00:41:59, Serial1/0
O E1     10.20.89.0/24 [110/84] via 172.21.11.11, 00:41:59, Serial1/0
O E1     10.20.97.0/24 [110/84] via 172.21.11.11, 00:41:59, Serial1/0
O IA     10.30.16.0/20 [110/66] via 172.21.12.2, 00:41:25, FastEthernet0/0
O E2     10.100.1.0/24 [110/20] via 172.21.11.11, 00:41:59, Serial1/0
      100.0.0.0/30 is subnetted, 1 subnets
O E2     100.100.102.0 [110/20] via 172.21.11.11, 00:41:59, Serial1/0
      172.20.0.0/16 is variably subnetted, 3 subnets, 3 masks
O E1     172.20.2.0/24 [110/84] via 172.21.11.11, 00:41:59, Serial1/0
O E2     172.20.11.0/28 [110/20] via 172.21.11.11, 00:41:59, Serial1/0
O E2     172.20.12.0/30 [110/20] via 172.21.11.11, 00:41:59, Serial1/0
      172.21.0.0/16 is variably subnetted, 4 subnets, 3 masks
C        172.21.11.0/28 is directly connected, Serial1/0
L        172.21.11.1/32 is directly connected, Serial1/0
C        172.21.12.0/30 is directly connected, FastEthernet0/0
L        172.21.12.1/32 is directly connected, FastEthernet0/0
      172.23.0.0/29 is subnetted, 1 subnets
O IA     172.23.23.0 [110/65] via 172.21.12.2, 00:41:25, FastEthernet0/0
      192.121.11.0/28 is subnetted, 1 subnets
O E2     192.121.11.0 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
      192.122.12.0/30 is subnetted, 1 subnets
O E2     192.122.12.0 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
      192.168.2.0/32 is subnetted, 1 subnets
O        192.168.2.1 [110/2] via 172.21.12.2, 00:41:25, FastEthernet0/0
      192.168.3.0/32 is subnetted, 1 subnets
O        192.168.3.1 [110/2] via 172.21.12.2, 00:41:25, FastEthernet0/0
      192.168.4.0/32 is subnetted, 1 subnets
O        192.168.4.1 [110/2] via 172.21.12.2, 00:41:25, FastEthernet0/0
O E2  192.168.10.0/24 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
O E2  192.168.11.0/24 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
O E2  192.168.20.0/24 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
O E2  192.168.30.0/24 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
O E2  200.100.20.0/24 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
```

```
O E2  200.100.20.0/24 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
      201.110.11.0/28 is subnetted, 1 subnets
O E2     201.110.11.0 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
      201.120.12.0/30 is subnetted, 1 subnets
O E2     201.120.12.0 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
      201.122.12.0/24 is variably subnetted, 2 subnets, 2 masks
C        201.122.12.0/30 is directly connected, Serial1/1
L        201.122.12.1/32 is directly connected, Serial1/1
      201.223.23.0/29 is subnetted, 1 subnets
O E2     201.223.23.0 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
      201.223.32.0/29 is subnetted, 1 subnets
O E2     201.223.32.0 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
      202.230.23.0/29 is subnetted, 1 subnets
O E2     202.230.23.0 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
O E2  203.23.30.0/24 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
O E2  210.11.1.0/24 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
O E2  210.11.11.0/24 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
O E2  210.200.20.0/24 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
O E2  220.202.20.0/24 [110/1] via 172.21.11.11, 00:11:08, Serial1/0
```

proper path resolution and end-to-end connectivity through the OSPF domain.

#5: Security mechanism and example with the configuration

BGP Security Mechanism, security mechanism are essential to prevent unauthorizes access, route hijacking and route leaks.

1. MD5 Authentication, to prevents unauthorized BGP session establishment
Prevets unauthorized BGP session establishment.

Example Configuration on both BGP peers (e.g. BG3 and ISP2):

router bgp 300
 neighbor 220.202.20.1 password cisco123

router bgp 200
 neighbor 220.202.20.2 password cisco123

2. Prefix Filtering (Route Filtering)
Limits what prefixes are accepted or advertised to avoid route hijacks

ip prefix-list ALLOW_ER2_ONLY seq 5 permit 172.20.2.0/24
 route-map FILTER_ER2_IN permit 10
 match ip address prefix-list ALLOW_ER2_ONLY

router bgp 300
 neighbor 220.202.20.1 route-map FILTER_ER2_IN in

Only allows route 172.20.2.0/24 from ISP2 to BG3.

3. Maximum Prefix, to prevents the router accepting too many routes from a peer.

```
router bgp 300
neighbor 220.202.20.1 maximum-prefix 50
```

if ISP2 sends more than 50 routes, the session is closed

4.Inbound ACL, Prevents unauthorized BGP session from untrusted IPs.

```
access-list 101 permit tcp host 220.202.20.1 host 203.23.30.2 eq 179
 access-list 101 deny tcp any any eq 179
 access-list 101 permit ip any any

interface GigabitEthernet0/0
 ip access-group 101 in
```

Only allows BGP TCP traffic from trusted neighbor.